

Navigating the confluence of artificial intelligence and education for sustainable development in the era of industry 4.0: Challenges, opportunities, and ethical dimensions

Ammar Abulibdeh ^{a,*}, Esmat Zaidan ^b, Rawan Abulibdeh ^c

^a Applied Geography and GIS Program, Department of Humanities, College of Arts and Sciences, Qatar University, P.O. Box: 2713, Doha, Qatar

^b College of Public Policy, Hamad Bin Khalifa University (HBKU), Qatar

^c Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada

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ABSTRACT

The emergence of Industry 4.0 marks a transformative era for businesses and industries, characterized by advanced technologies like automation, Internet of Things (IoT), artificial intelligence (AI), smart factories, and cyber-physical systems. This revolution promises significant advantages, including enhanced productivity, sustainable progress, and heightened resilience. However, the integration of Industry 4.0 is challenged by the need for a skilled workforce with expertise in areas such as information technology and data analytics. Higher education institutions (HEIs) play a vital role in equipping future professionals with these skills, necessitating curriculum updates and infrastructure enhancements. Simultaneously, the importance of education for sustainable development (ESD) has been underscored by global initiatives like the Sustainable Development Goals (SDGs). ESD instills a sense of responsibility for economic, ecological, and equitable well-being. As digital technologies blur the lines between industries, education faces the challenge of adapting to evolving demands. The integration of AI tools in education has emerged as a catalyst for reshaping learning experiences, fostering innovation, and preparing individuals for the digital age. AI chatbots such as ChatGPT have garnered widespread attention and possess the potential to revolutionize various aspects of education. However, their integration raises ethical concerns, necessitates curriculum redesign, requires strategies for continuous learning, and demands alignment with industry standards. While the potential of AI integration in education is promising, there is a notable gap in the existing literature when it comes to exploring the ethical implications, the influence of AI on ESD, the impact on the structure of Blooms Taxonomy, collaboration between academia and industry, strategies for continuous learning, and the effective integration of AI tools for personalized learning. This paper aims to critically examine the integration of AI tools, with a specific emphasis on ChatGPT, in education within the context of ESD. It delves into the transformative potential, ethical considerations, imperatives for continuous learning, and the role of industry partnerships. By providing insights and strategies, this paper contributes to the ongoing discussion about the evolving nature of education in a technologically driven world, equipping academic institutions to navigate the complexities and opportunities associated with AI integration in education more effectively.

1. Introduction

The advent of the fourth industrial revolution (Industry 4.0) has induced a profound metamorphosis in the operational and evolutionary dynamics of businesses and industrial activities. Characterized by an array of state-of-the-art technologies including automation, system integration, decentralization, the Internet of Things (IoT), Artificial Intelligence (AI), smart factories, simulation, data analytics, cloud

manufacturing, big data, and cyber-physical systems, among others (Brahma et al., 2021; Fuertes et al., 2021; Hammad Mian et al., 2020; Jan et al., 2023), this industrial revolution heralds manifold advantages. These technologies delineate the analogous attributes inherent to the Industry 4.0 epoch. Furthermore, the incorporation of these attributes holds the potential for expediting enterprise expansion, heightened productivity, enhanced process management and orchestration, sustainable progress, augmented resilience, and other notable outcomes.

* Corresponding author.

E-mail address: aabulibdeh@qu.edu.qa (A. Abulibdeh).

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The transition from Industry 3.0 to Industry 4.0 is intrinsically geared towards achieving digitalization, wherein a virtual domain is engendered by the seamless integration of digital, physical, and biological systems (Aoun et al., 2021; Caiado et al., 2021; Pascoal-Faria et al., 2023).

The realization and integration of Industry 4.0 are subject to limitations arising from the demand for individualized proficiencies and expertise within the workforce (Aoun et al., 2021). This necessitates the cultivation of adaptive and analytical thinking capabilities, particularly focused on domains such as information technology and data analytics. Higher education institutions (HEIs) stand as pivotal entities in equipping the forthcoming workforce with these proficiencies (Leow et al., 2023; Musariwa and Tinonetsana, 2023). Societies are consequently compelled to contemporize their educational curricula, infrastructural resources, and facilities, aligning them with the evolving landscape of Industry 4.0. This endeavor embodies both prospects and complexities as Industry 4.0 unfolds its transformative potential (Amaral and Peças, 2021). Certainly, a myriad of constituent elements necessitates a judicious evaluation for the purpose of orchestrating this transformative process strategically. Undoubtedly, this inquiry underscores the significance of hands-on proficiency and the integration of digital technologies within the university domain. This integration serves to equip emerging talents with essential aptitudes and a competitive edge tailored for Industry 4.0.

The United Nations Conference on Environment and Development, convened in Rio de Janeiro in 1992, underscored the paramount significance of higher education and pedagogy in effectuating the tenets of sustainability enshrined in the Sustainable Development Goals (SDGs). This sentiment was further accentuated when the UN designated the span from 2005 to 2014 as the Decade of Education for Sustainable Development (ESD) (Piroșcă et al., 2020). The nomenclatures “sustainable education,” “education for sustainable development,” and “education for sustainability” encapsulate the learning process wherein deliberations revolve around the enduring well-being of economic, ecological, and equitable facets of communities (Lukman and Glavić, 2007; Neamțu et al., 2020; Șerban et al., 2020). ESD has emerged as a nascent yet dynamic paradigm, furnishing a comprehensive framework for instilling a sense of responsibility across citizens of all age strata, encouraging proactive participation in the shaping of a sustainable future consonant with SDGs (Neamțu et al., 2020). This form of education assumes critical relevance in operationalizing the principles of sustainable development (SD), furnishing participants with the acumen to make well-informed choices, cognizant of their profound influence and consequential obligations in steering future societal progress. Concomitantly, education imbued with the principles of SD impels participants to act judiciously, ensuring environmental robustness, economic viability, and the creation of a secure milieu for current and succeeding generations.

As digital technologies seamlessly blend with physical systems and redefine the boundaries of industries, the landscape of education finds itself at a crossroads. Traditional educational paradigms are being challenged by the imperative to adapt and evolve in the face of rapidly changing demands and opportunities (Sok and Heng, 2023). In this dynamic context, the integration of AI tools into education has emerged as a key catalyst for reshaping learning experiences, fostering innovation, and preparing individuals for the complexities of the digital age (Wölfel and Taecharungroj, 2023). The intersection of AI and education presents a paradigm shift that holds promises and complexities alike. AI tools, such as ChatGPT and other advanced language models, have demonstrated their potential to revolutionize various aspects of education – from content creation and personalization to interactive engagement and beyond (Baidoo-Anu and Owusu Ansah, 2023).

ChatGPT, an AI chatbot employing natural language processing for responding to queries, has rapidly garnered significant attention within contemporary culture (Day, 2023; Rudolph and Tan, 2023; Thorp, 2023). This emergence has engendered substantial ramifications across

both scientific and societal domains (van Dis et al., 2023). Its swift embrace and comprehensive exploration within the broader population have instigated an array of extensive dialogues concerning its suitable and optimal assimilation across diverse domains. ChatGPT represents the inaugural instance of a substantial large language model (LLM) to attain widespread recognition among the general populace. LLMs have showcased notable proficiency (and at times limitations) in generating text in response to prompts. Certain LLMs, such as Elicit and Med-PaLM, and Tiangong AI, Kaiwu, possess the capacity to scour existing literature and propose specific inquiries or insights pertaining to a given topic or question by harnessing available knowledge (Biswas, 2023; Fuchs, 2023). The latest iteration, GPT-4, has also acquired the capability to assimilate information from images, thereby amplifying the potential applications of LLMs, particularly within education, healthcare, and research domains where visual depictions are pivotal for fostering comprehension or augmenting insights (Agathokleous et al., 2023). However, as these tools become more prevalent in educational settings, critical considerations surrounding ethics, curriculum design, continuous learning, and industry alignment come to the forefront.

The integration of AI tools into education within the context of Industry 4.0 has been receiving growing attention, primarily because of its capacity to redefine traditional educational paradigms (Mourtzis et al., 2022; Schiele et al., 2022). However, upon a critical review of the existing literature, several significant gaps emerge, highlighting the need for in-depth exploration and analysis. For instance, while AI integration in education holds immense promise, there is a lack of comprehensive exploration of the ethical implications surrounding its adoption (Rodríguez-Amigo et al., 2022). The existing literature predominantly concentrates on the technical and pedagogical aspects, often overlooking the influence of AI on ESD. This includes ethical dimensions linked to AI-generated content, students' critical thinking, teacher and student competencies, data privacy concerns, and potential biases. Furthermore, despite the acknowledged importance of aligning education with the evolving demands of Industry 4.0, the research on how academic institutions can effectively collaborate with industries to bridge the education-industry divide remains limited. The available literature lacks comprehensive insights into successful collaboration models and the verification of their impact on preparing students for the digital age. Additionally, the rapidly changing technological landscape of Industry 4.0 necessitates a continuous learning mindset. However, there is a gap in the literature when it comes to effective strategies for implementing continuous learning within educational institutions. Many studies discuss the importance of lifelong learning without providing practical approaches for integrating it into curricula and teaching methodologies. Lastly, while some studies acknowledge the potential of AI in transforming pedagogical practices, there is a lack of comprehensive research on how AI-powered tools can be effectively integrated into the classroom to enhance personalized learning experiences. The existing literature often lacks in-depth analysis of AI's impact on pedagogy beyond surface-level discussions.

This paper aims to provide a comprehensive and critical analysis of the integration of AI tools in education within the context of ESD. By illuminating the potential for transformation, addressing ethical concerns, emphasizing the need for continuous learning, and highlighting the significance of industry partnerships, this paper enriches the ongoing conversation about the evolving landscape of education in an interconnected and technologically driven world. Through an in-depth exploration of research findings, this paper intends to empower academic institutions with the insights and strategies essential for effectively navigating the complexities and opportunities arising from the integration of AI in education.

2. Methodology

A scoping review focusing on the integration of AI tools in education, within the context of ESD, was conducted as the primary method for this

study. This approach provided a comprehensive overview of the existing literature at the intersection of AI and ESD in the era of Industry 4.0. Unlike systematic reviews, which concentrate on a specific research question, scoping reviews adopt a broader scope (Peters et al., 2015; Pham et al., 2014). They aim to map out the key concepts underlying a research area, as well as the main sources and types of evidence available. Additionally, scoping reviews are instrumental in identifying gaps in the literature (Thomas et al., 2019). These methods seek to provide a comprehensive overview of the evidence, rather than a quantitative or qualitative synthesis of data. This makes scoping reviews particularly suitable for topics that are emerging or not yet fully established.

2.1. Search strategy

References for this review were identified through searches in Web of Science, Scopus, and Google Scholar. The inclusion and exclusion process adhered to the standards set by the PRISMA extension for scoping reviews (Tricco et al., 2018). Relevant articles were initially identified by reading their titles and abstracts, with a subsequent in-depth review of the full papers. Keywords and controlled vocabulary terms, established through literature searches and domain expertise, were used in

the searches. These included combinations of terms related to Industry 4.0, education, and AI, as illustrated in Fig. 1. Additional keywords were derived from the abstracts of previously included articles. The initial search yielded 384 studies, from which 57 duplicates were removed. However, this search uncovered a limited number of relevant articles, reflecting the novelty of this topic in the context of recent advancements in AI and the limited studies indexed by databases such as Scopus and Web of Science. Consequently, a snowball technique was employed to identify additional studies. We also manually searched other resources, including IEEE Xplore, to find further relevant articles.

2.2. Study selection

The inclusion criteria for this review were as follows: (a) original research articles; (b) published within the last 20 years (2003–2023); (c) published in English; and (d) alignment with the conceptual framework of this study. The exclusion criteria were: (a) articles not published in English; (b) commentaries or editorials; and (c) other review articles. All papers underwent a two-stage verification process by two independent reviewers to be included in this study, starting with screening the title and abstract, followed by a full-text review. Any disagreements

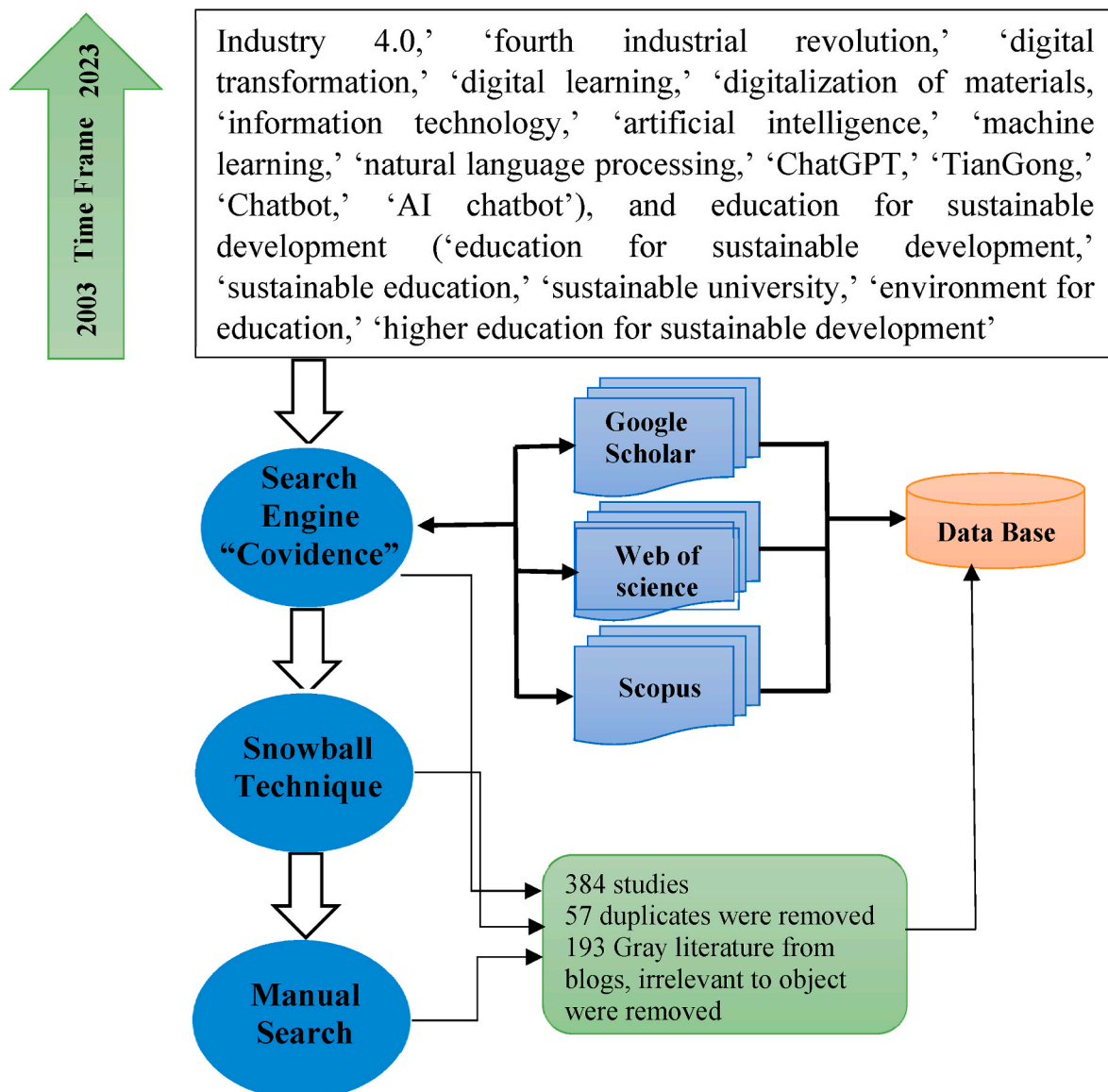


Fig. 1. Flowchart of literature search for scoping review on the intersection of AI and ESD in the era of industry 4.0.

regarding study selection were resolved through consensus and discussion with a third reviewer. Covidence, a web-based software platform for screening and data extraction, facilitated the conduct of this literature review. The final selection comprised 134 peer-reviewed articles, including contributions from different journals, conference proceedings, book chapters, and institutional reports that met our eligibility criteria. Most of the reviewed articles were published in the last three years, indicating a recent surge in research. Future advancements in AI within the education sector may pave the way for a more comprehensive systematic review of the literature.

2.3. Methodological framework

Studying the impact of AI on ESD is a multifaceted undertaking, demanding a comprehensive framework that encompasses various dimensions of the subject. To conduct this research effectively, it is crucial to follow a structured approach. The suggested framework (see Fig. 2) consists of verity of interconnected steps, beginning with the definition of key concepts and objectives. These encompass understanding the principles of ESD, exploring the role of AI in education, and the role of HEIs. A literature review was conducted to identify existing knowledge

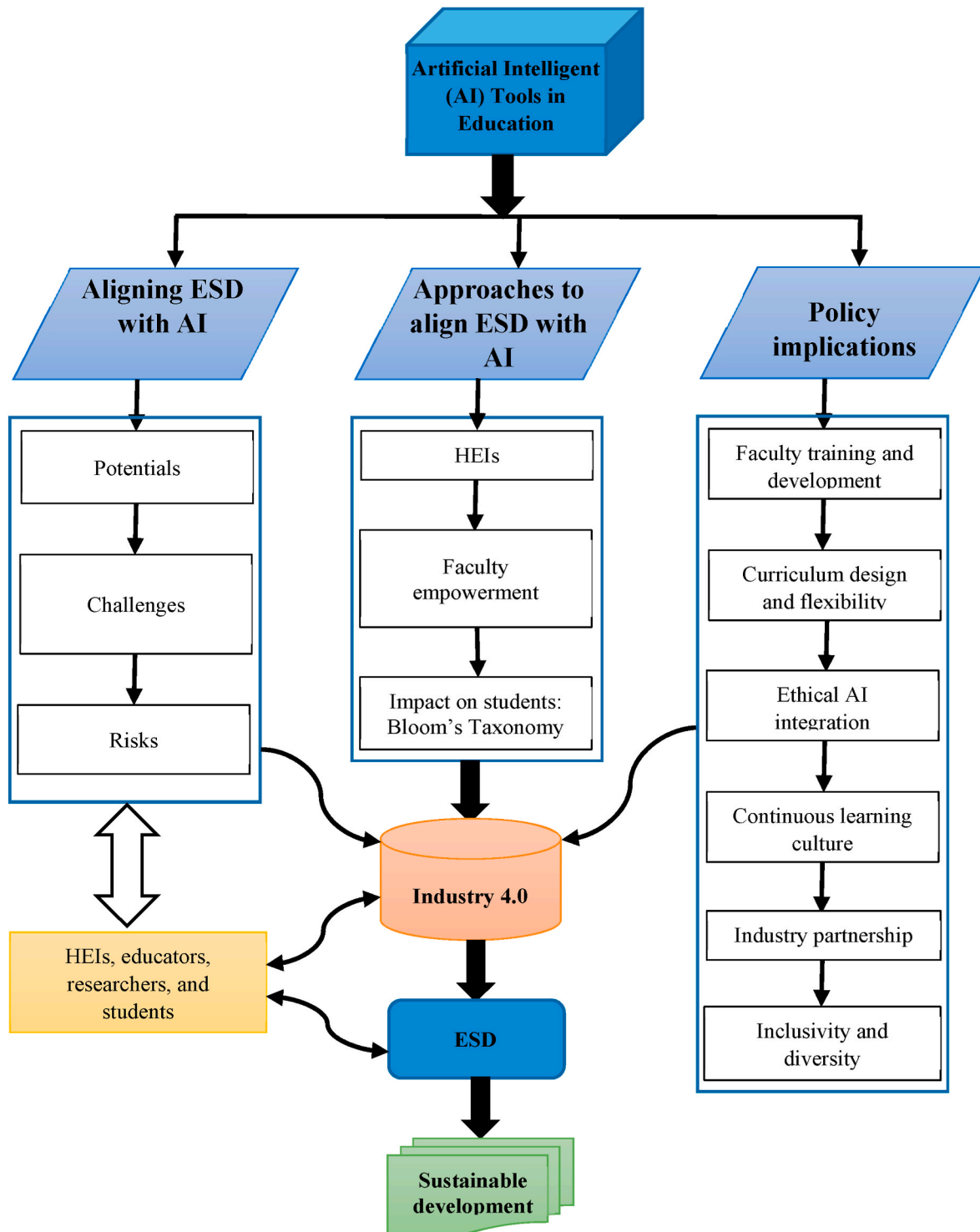


Fig. 2. Framework of the study.

and research gaps. The subsequent steps delve into AI technologies and their potential, challenges, and risks in education, examining their impact on teaching and learning, curriculum development, data privacy, and student outcomes. The framework extends further to address different approaches to align ESD with AI including institutional and policy implications, ethical considerations, and the bridging the education-industry divide in the Era of Industry 4.0. Ultimately, the research culminates in results and discussion sections, and concluding with the summarization of key findings and their significance for the field. This structured approach ensures a thorough investigation of AI's role in advancing ESD, encompassing technological, educational, ethical, and policy perspectives.

3. Results

3.1. Reimagining education for sustainability in the era of industry 4.0: nurturing innovators for a sustainable and technologically advanced future

ESD revolves around the establishment of an institutional “ecosystem,” meticulously crafted to furnish swift and efficient support services capable of promptly addressing the exigencies of students and employers pertaining to the application of sustainability principles across diverse domains and activities (Baker-Shelley et al., 2017; Paletta et al., 2019; Staniškis and Katiliute, 2016). The challenge in executing ESD resides in cultivating a pedagogical and didactic framework that accommodates this specific form of student training and the impartation of pertinent competencies (Disterheft et al., 2015). To this end, universities are compelled to reevaluate their academic curricula, aligning them more adeptly with the requisites of the labor market while remaining attuned to the training requisites articulated by employers (Plaia et al., 2011). This entails the integration of elements that facilitate both the enduring solidification of taught material (Jora et al., 2020) and the dissemination of public consciousness regarding the notion of SD, its manifold dimensions, and core tenets, fostering a mindset favorable to the concept.

Equally imperative is the comprehensive instruction of academic personnel concerning a precise comprehension of ESD and the adept transference of its tenets into instructional subjects (García-González et al., 2020), with a particular emphasis on its seamless incorporation within the educational curricula (Jora et al., 2020). The cultivation of a mindset predisposed toward embracing the construct of SD will not only bestow students with an enhanced apprehension of the sustainability paradigm and its practical applications, but also foster a sense of conscientious development, capacitating them to emerge as veritable champions of sustainable transformation in both their individual and professional realms (Mulà et al., 2017; Piroșcă et al., 2020). This ethos would further empower students to make judicious decisions and actively partake in tangible initiatives within the overarching backdrop of societal, economic, and environmental challenges (Annan-Diab and Molinari, 2017; Casarejos et al., 2017).

Reimagining ESD is a pressing need as societies navigate the era of Industry 4.0. In this transformative industrial era, HEIs need to adapt to align with the demands of the rapidly evolving job market and the imperative for sustainable practices (AlMalki and Durugbo, 2023; Hammad Mian et al., 2020). A holistic approach to integrating sustainability principles in higher education can be a powerful catalyst for preparing students to thrive in the Industry 4.0 landscape.

The Industry 4.0 era emphasizes the importance of innovation and adaptability. ESD fosters critical thinking, problem-solving, and a holistic understanding of complex global challenges, all of which are essential skills in the digital age (Kalsoom and Khanam, 2017). By infusing sustainability principles throughout the curriculum, students are encouraged to think creatively and develop solutions that not only address environmental and social issues but also align with the needs of technologically advanced industries. This alignment ensures that graduates possess the interdisciplinary skills required to contribute

meaningfully to Industry 4.0 sectors (Baena-Morales and Fröberg, 2023; Hensley, 2020). For example, Baena-Morales and Fröberg (2023) emphasize the importance of a learning environment that encourages exploration, reflection, and environmental problem solving. They suggest that university students have the capacity to analyze and understand more advanced and complex concepts, and that sustainability can be taught in relation to their area of study, fostering the development of specific and professionally applicable skills.

Furthermore, Industry 4.0 emphasizes data-driven decision-making and the utilization of advanced technologies. SD principles underscore the responsible use of resources and ethical considerations, which are critical components in the ethical and sustainable implementation of Industry 4.0 technologies (Ching et al., 2022; Verma et al., 2022). HEIs that integrate sustainability into their programs can instill an ethical mindset among students, ensuring that they apply emerging technologies with a focus on long-term environmental and societal impacts (Tasdemir and Gazo, 2020). Gupta et al. (2021) discuss the principle of Industry 4.0, which involves the real-time exchange of data and flexible manufacturing systems. They argue that this principle can contribute to sustainability by improving environmental quality and promoting innovative business initiatives that align with environmental, societal, and economic goals. Lastly, Industry 4.0 thrives on collaboration and interconnectedness. ESD often emphasizes interdisciplinary teamwork and engagement with stakeholders from various sectors. By fostering collaboration and a systems-thinking approach, HEIs can prepare students to work effectively in the interconnected web of Industry 4.0, where collaboration between industries, academia, and society is key to addressing complex challenges (Bildirci and Ersin, 2023; Khan et al., 2023). For example, Khan et al. (2023) evaluate the implications of Industry 4.0 innovations from a SD perspective, underscoring the importance of integrating sustainability and SD paradigms into Industry 4.0. In this way, a holistic approach to ESD not only aligns with Industry 4.0 but also equips students with the skills and mindset to thrive in this era of technological transformation.

3.2. Potentials, challenges, and risks of aligning ESD with AI tools to meet the demands of industry 4.0

In the rapidly evolving landscape of Industry 4.0, the alignment of ESD with AI tools has become imperative. This alignment serves as a bridge between traditional curricula and the dynamic demands of modern industries. However, it presents significant formidable challenges and risks. On the one hand, the integration of AI-powered technologies into educational frameworks has the potential to revolutionize the way students are prepared for a sustainable future in a rapidly evolving industrial landscape. AI can provide personalized learning experiences, real-time data analysis, and data-driven insights into industry trends and sustainability challenges. This can lead to more relevant and effective educational programs that equip students with the skills and knowledge needed to tackle the complex problems of sustainability within the context of Industry 4.0. However, on the other hand, there are substantial challenges and risks to navigate, including issues related to access, bias, privacy, overreliance, and resistance to change. Striking the right balance between harnessing the potentials of AI for SD and addressing these challenges is essential to ensure that AI-enhanced education effectively prepares students for the demands of Industry 4.0 while promoting ethical, inclusive, and equitable education for a more sustainable future.

3.2.1. Unlocking the potential of AI in aligning ESD with industry 4.0

The integration of ESD with AI tools in HEIs can potentially revolutionize the way we approach education in the era of Industry 4.0. This alignment enables HEIs to offer more relevant and up-to-date curricula and enhance the delivery of ESD content by providing personalized and adaptive learning experiences (Grimus, 2020). AI tools can aid in analyzing industry trends, job market demands, and sustainability

challenges in real time, providing invaluable insights to educators (Dwivedi et al., 2021). With this information, institutions can adapt and tailor their programs to ensure that graduates possess the latest skills and knowledge demanded by rapidly evolving industries (Dlouhá et al., 2017). This approach can lead to more extensive changes in teaching and learning processes, fostering innovation in university curricula. This dynamic responsiveness not only enhances the quality of education but also helps students graduate with a competitive edge, ready to contribute meaningfully to the ever-changing job market.

The use of AI tools can facilitate the transdisciplinary character of ESD initiatives (Kubisch et al., 2020). Sustainability challenges often require multifaceted solutions that involve input from various disciplines. By fostering cooperation between actors and entities within and outside academia, AI can help in creating a more holistic and integrated approach to ESD (Beynaghi et al., 2016; Fekih Zguir et al., 2021). This can lead to the development of critical and reflective professionals who not only possess the appropriate knowledge in their discipline but also understand how their actions can contribute to the construction of more sustainable societies (Sammalisto et al., 2015; Sánchez-Carracedo et al., 2021). Aligning education with AI tools to meet the demands of Industry 4.0 requires a comprehensive approach that includes curriculum development, pedagogical strategies, collaboration with industry partners, and the integration of emerging technologies (Brahma et al., 2021). By adopting these approaches, educational institutions can prepare students to thrive in the era of technological transformation and contribute meaningfully to Industry 4.0 sectors.

Another significant potential is the enhancement of skills relevant to Industry 4.0. AI can assist in identifying emerging job roles and skill requirements within various industries (Dlouhá et al., 2017; Fekih Zguir et al., 2021). By constantly analyzing industry trends and job market data, educational institutions can adapt their curricula to ensure students are equipped with the most up-to-date skills and competencies (Mourtzis et al., 2022). This proactive approach not only benefits students but also strengthens the connection between academia and industry, fostering greater collaboration and relevance. As AI tools facilitate the development of skills like data analysis, problem-solving, and digital literacy, students become better prepared to navigate the technological advancements of Industry 4.0 and contribute meaningfully to their chosen fields (Kar et al., 2022). Furthermore, the potential for lifelong learning is expanded through AI-powered education. Industry 4.0 is characterized by continuous technological advancements, requiring professionals to adapt and acquire new skills throughout their careers. AI-driven educational platforms can offer flexible and accessible learning opportunities, enabling individuals to upskill and reskill as needed. Whether through online courses, micro-credentials, or personalized learning pathways, AI-supported education can empower individuals to stay competitive in the evolving job market, fostering a culture of lifelong learning that is essential in Industry 4.0.

AI can play an important role in connecting students from diverse academic backgrounds, encouraging collaboration on projects that address real-world sustainability issues. This approach equips students with crucial teamwork, problem-solving, and communication skills, making them well-prepared for the collaborative nature of work in Industry 4.0 (Kar et al., 2022). The potential for HEIs to foster a multi-disciplinary approach through AI-driven ESD not only enhances students' learning experiences but also equips them with the skills needed to navigate the complex, interconnected challenges of the modern workforce (Klimova et al., 2016; Walter Leal Filho et al., 2021). The integration of AI tools in ESD fosters personalized learning experiences. Each student has unique learning styles, preferences, and strengths. AI algorithms can analyze individual data to create customized learning pathways, adjusting content and pace accordingly (Walter Leal Filho et al., 2021). This personalized approach enhances student engagement and comprehension, catering to their specific needs and ensuring a more effective learning process. By harnessing AI's potential for personalization, HEIs can maximize the efficiency of education

delivery, ultimately benefiting both students and educators.

3.2.2. The challenges of integrating AI into ESD for industry 4.0

However, there are several significant challenges associated with aligning ESD with AI tools in the context of Industry 4.0. One of the foremost challenges is the digital divide (Lambrechts et al., 2013). Not all students have equal access to technology and the internet, which can exacerbate educational inequalities. Overreliance on AI tools can inadvertently exclude students who lack access to these resources, creating disparities in learning opportunities (Dinca et al., 2007; Stephens et al., 2008). Bridging the digital divide and ensuring equitable access to AI-driven education is a critical challenge that educational institutions and policymakers must address. Another challenge is the potential for AI bias and ethical concerns (Klimova et al., 2016). AI algorithms can inherit biases from the data they are trained on, leading to unfair and discriminatory outcomes. In the context of ESD, biased AI tools could provide skewed recommendations for sustainability initiatives or perpetuate existing inequalities related to environmental justice (Walter Leal Filho et al., 2021). Safeguarding against AI bias and ensuring ethical AI development and deployment in education are paramount to maintaining fairness and inclusivity. The rapid pace of technological change poses a challenge. Industry 4.0 is characterized by constant innovation, making it difficult for educational institutions to keep their curricula and AI tools up to date (Dlouhá et al., 2017). This is particularly true for ESD, which requires a holistic and integrated approach to education (Fekih Zguir et al., 2021; Sánchez-Carracedo et al., 2021). Outdated content and tools can quickly become obsolete, leaving graduates ill-prepared for the evolving sustainability challenges of the Industry 4.0 era (Sammalisto et al., 2015). Continuous professional development for educators and agile, adaptive AI systems are essential to address this challenge (Ahmad et al., 2023). Furthermore, there is the challenge of resistance to change within educational institutions. Implementing AI-driven ESD requires significant changes in curriculum design, teaching methods, and institutional practices (Sharma et al., 2022). Resistance from educators, administrators, and other stakeholders can impede the adoption of these innovative approaches. Overcoming this resistance and fostering a culture of innovation and adaptability within educational institutions is crucial to harnessing the full potential of AI for SD in Industry 4.0.

3.2.3. The risks of integrating AI into ESD for industry 4.0

Privacy and data security issues are significant risks when integrating AI into ESD for Industry 4.0 (Dilmaghani et al., 2019; Kar et al., 2022; Li and Zhang, 2017). AI tools in education often collect and analyze sensitive student data. Mishandling or data breaches can have severe consequences, including violations of student privacy and ethical concerns. Educational institutions should implement robust data protection measures, adhere to data privacy regulations, and educate students about their data rights to safeguard against these risks. Furthermore, there is the risk of overreliance on AI tools (Buçinca et al., 2021; Gaube et al., 2021). While AI can enhance education, it should not replace the critical role of educators in guiding and mentoring students. Excessive automation and depersonalization of education can lead to a loss of the human touch and critical thinking skills (Tokayev, 2023). Striking the right balance between AI-driven learning and human interaction is essential to ensure a holistic and effective educational experience that prepares students for the challenges and opportunities of Industry 4.0 and SD. There is also the risk of reducing sustainability to a technical exercise. AI tools can provide data-driven solutions to sustainability challenges, but they should not overshadow the importance of social, cultural, and ethical dimensions of sustainability (Del Río Castro et al., 2021). The risk lies in focusing solely on the technical aspects of sustainability and neglecting the broader context and human values that underpin SD. Another risk is the potential for AI-driven education to become overly standardized (Bibri et al., 2024). While personalization is a significant benefit, there is a risk that AI tools may

push education toward a one-size-fits-all approach if not carefully managed. Customization should not result in a loss of diversity in educational approaches and perspectives, as diversity is essential for holistic education.

3.3. Approaches to aligning ESD with AI tools to meet the demands of industry 4.0

In today's rapidly evolving world, the integration of technology and sustainability has become paramount. Simultaneously, the imperative for SD has grown more pressing than ever as societies grapple with complex challenges such as climate change, resource scarcity, and social equity. To address these multifaceted issues and prepare the workforce for Industry 4.0, it is crucial to align ESD with the capabilities of AI. This convergence presents transformative potential, enabling educational institutions to equip students with the knowledge, skills, and mindset needed to thrive in a world where sustainability and technology are deeply intertwined. This section highlights different approaches that can be considered to align ESD with AI tools to meet the demands of Industry 4.0.

3.3.1. HEIs in the industry 4.0 era: sustainable education, AI integration, and societal transformation

Throughout history, HEIs have fulfilled a pivotal role in societal transformation, being responsible for the education of decision-makers, leaders, entrepreneurs, and scholars (Cortese, 2003; Elton, 2003; Lozano, 2006; Lozano et al., 2013), and advocating for the greater public welfare (Mulder, 2010; Waas et al., 2010). HEIs assume a pivotal role in cultivating a pro-sustainability mindset by seamlessly integrating pedagogy, institutional operations, and research endeavors in symbiotic synergy with external communities (Hugé et al., 2016; Kemper et al., 2020). The existing literature underscores that a sustainable university's foundation rests upon the synergistic interplay of three primary vectors: the internal facet, concentrating on the university campus and its constituents; the external dimension, directed at the community and the relationships forged with socio-economic collaborators, notably students, and their cultivation of knowledge, skills, and competencies; and the inculcation of sustainability principles into them (Argento et al., 2020; Pocol et al., 2022; Șerban et al., 2020; Valentin and Nagore, 2014).

Over the past twenty years, an escalating number of HEIs have undertaken endeavors to integrate and institutionalize the principles of SD into their frameworks. Various HEIs have embarked on this journey with the intent to make substantial progress in this domain (Lozano et al., 2013). However, it is noteworthy that, in various aspects, universities have remained rooted in traditional practices (Elton, 2003), contributing to and even hastening unsustainable developmental trajectories (Wals, 2008), and resisting transformative shifts. Consequently, the perpetuation of Newtonian and Cartesian paradigms has been observed, and while these approaches have proven beneficial for societies in numerous cases, in others they have fostered a pronounced emphasis on the mastery of nature and the industrialization of the planet. This, in turn, has resulted in the production of graduates who are over-specialized, mono-disciplinary, and imbalanced in their perspectives (Cortese, 2003; Lozano et al., 2013). This scenario presents a formidable predicament for HEIs and society alike, particularly in light of the sluggish pace of change exhibited by entities such as universities.

Compared to the educational landscape of the Twentieth Century, which predominantly revolved around skill transmission and knowledge consolidation, Twenty-First Century education is distinctly centered on nurturing creativity, fostering collaborative engagement among students within the pedagogical journey, and notably, cultivating "soft skills" that are pivotal attributes conducive to enriching students' erudition (Burnard, 2006; Sanina et al., 2020). Contemporary higher education curricula necessitate the integration of pedagogical tools based on digital simulations, the establishment of practice-oriented

communities, and the facilitation of joint-teaching sessions involving both domain specialists and instructional personnel (Stanca et al., 2022). These approaches not only foster the co-creation of knowledge but also expedite the assimilation of entrepreneurial competencies within students. Collaborative knowledge generation at the intersection of academia and the business environment yields an educational product that seamlessly integrates into students' learning and practical experience (Han, 2020; Stanca et al., 2022). Pigozzi (2007) posited that contemporary educational paradigms should align with the requisites of the contemporary global landscape, encompassing not only cognitive aspects but also embracing social and additional dimensions of learning. Laurie et al. (2016) expounded upon the notion that an all-encompassing strategy fosters a more expansive acquisition of knowledge, resulting in the attainment of curriculum objectives. Furthermore, they emphasized the indispensability of comprehensive teacher development, essential for upholding educational excellence, executing ESD policies, and facilitating multidimensional student learning. Ambusaidi and Al Washahi (2016) underscore the paramount significance of education in the attainment of SD, underscoring the imperative of a holistic reconfiguration of attitudes and behaviors across diverse domains, a feat feasible only through education's proactive involvement in the cultivation of individual and societal dispositions and behaviors.

The role of HEIs in harnessing AI tools for learning is pivotal in shaping the future of education. These institutions serve as both custodians of knowledge and pioneers of innovation, making them uniquely positioned to leverage AI's transformative potential. However, these institutions must address ethical considerations and guide students in using AI tools, such as ChatGPT responsibly, ensuring that this AI-driven tool augments, rather than replaces, critical thinking and problem-solving skills.

The global higher education sector is characterized by a diverse range of contextual complexities and nuances, shaped by various factors like cultural, economic, political, and technological influences (Shah-jahan et al., 2022). HEIs around the world cater to an increasingly diverse student population (Gill and Singh, 2020). This diversity brings a wealth of cultural perspectives but also challenges in ensuring inclusivity and equity. Universities intend to navigate language barriers, cultural differences, and varied educational backgrounds, all while striving to create an inclusive learning environment (Bovill et al., 2016). The rapid advancement of technology has transformed higher education, introducing online learning, digital resources, and AI-driven educational tools (Bonfield et al., 2020). However, this shift also highlights the digital divide, as students in less developed regions or from disadvantaged backgrounds may lack access to necessary technology and internet connectivity. Furthermore, economic factors greatly influence the accessibility of higher education (Tchamyou, 2020). In some regions, higher education remains a privilege of the affluent, while in others, there are significant efforts to make it more accessible to all socioeconomic groups. Financial challenges, such as tuition fees and the availability of scholarships, grants, and student loans, vary widely between countries and institutions (Tchamyou, 2020). HEIs are increasingly operating in a globalized context, with international collaborations, student and faculty exchanges, and cross-border research initiatives (Romani-Dias et al., 2019). This internationalization brings opportunities for global learning and exposure but also challenges in balancing local priorities with global trends and standards. HEIs are increasingly expected to address global challenges such as climate change, sustainability, and social justice. Integrating these issues into the curriculum and research, while also managing operational sustainability, presents complex challenges.

Aligning the complexities and nuances of the global higher education sector with ESD in the era of Industry 4.0 involves several key considerations. The diverse factors shaping HEIs reflect the core principles of ESD, which emphasize the need to understand and address interconnected global challenges (Cebrián et al., 2021). ESD in Industry 4.0

should account for these diverse influences, integrating cultural sensitivity, economic viability, political awareness, and technological advancements into educational programs. The challenge of catering to a diverse student body aligns with ESD's goal of promoting inclusive and equitable quality education. In the context of Industry 4.0, this means not only addressing cultural and linguistic diversity but also equipping students with the skills to navigate a technologically advanced and interconnected world. Furthermore, the rapid integration of technology in higher education underscores the need for ESD to focus on reducing the digital divide. Ensuring equitable access to digital resources and AI-driven educational tools is crucial for preparing all students for the demands of Industry 4.0. The varying economic factors influencing access to higher education highlight the need for ESD to incorporate economic sustainability and social equity. Strategies to make higher education more accessible align with ESD's commitment to lifelong learning opportunities for all.

These complexities demonstrate that higher education is not just about imparting knowledge; it is a multifaceted sector influenced by a myriad of social, economic, political, and technological factors. The complexities of the higher education sector present both challenges and opportunities for aligning with ESD in the era of Industry 4.0. A holistic and adaptive approach is required to navigate these complexities, ensuring that education not only imparts knowledge but also fosters SD, inclusivity, and responsible citizenship in an increasingly interconnected and technologically advanced world.

3.3.2. Holistic curriculum design for industry 4.0: integrating AI tools and cultivating essential skills

As the world transitions into the era of Industry 4.0, the landscape of education is undergoing a seismic shift (Brahma et al., 2021; Mhlana, 2023; Mogali, 2023). Traditional paradigms of learning are being redefined to meet the demands of a rapidly evolving job market. In this transformative context, the role of academic institutions extends beyond imparting technical knowledge; it necessitates a comprehensive overhaul of curriculum design (Ihsan et al., 2023; Mampane, 2020). The integration of AI tools like ChatGPT becomes not only an opportunity but also a necessity to prepare students for the multifaceted challenges and opportunities of Industry 4.0.

In the context of Industry 4.0, the traditional silos of knowledge are no longer sufficient. A holistic curriculum design becomes imperative to equip students with the multidisciplinary skills needed to thrive in this dynamic landscape (Gordijn and Have, 2023). The integration of AI tools like ChatGPT offers a gateway to facilitate this integration. These tools can serve as supplementary resources that enrich students' learning experiences by providing real-time information, insights, and diverse perspectives from across disciplines (Agathokleous et al., 2023; Xu et al., 2018). This not only fosters a broader understanding of concepts but also nurtures curiosity and an interdisciplinary mindset, which are critical for innovation and problem-solving.

Courses should be designed to encourage critical thinking and problem-solving by presenting complex, open-ended challenges that necessitate analytical reasoning (Brahma et al., 2021). AI tools can be harnessed to provide data and insights that inform these challenges (King, 2023), prompting students to apply their technical knowledge in novel and innovative ways. Similarly, integrating AI tools into creative disciplines can provide inspiration, varied perspectives, and opportunities for experimentation, fostering a culture of innovation (Mhlana, 2023; Mogali, 2023). Prioritizing the development of emotional intelligence and adaptability in the curriculum is essential, as these intrinsically human skills are irreplaceable by AI and critical in the modern workforce. Emotional intelligence, which encompasses empathy, self-awareness, and emotional management, is vital for effective communication, leadership, and collaboration in diverse professional settings, areas where AI falls short. Adaptability equips individuals to navigate and thrive amidst rapid technological and societal changes, promoting resilience and innovative thinking. However, some argue this

focus might divert attention from equally important technical skills or theoretical knowledge, essential in an increasingly technology-driven world. Balancing these human-centric skills with technical proficiency is crucial, ensuring a well-rounded educational approach that prepares individuals for both current and future challenges.

Collaborative projects, interactive discussions, and role-playing scenarios can be used to simulate real-world situations where students should navigate interpersonal dynamics and uncertainties, thereby enhancing their emotional resilience and adaptability (Cebrián et al., 2020; Geithner and Menzel, 2016). However, the integration of AI tools into the curriculum also necessitates a cautious approach. It is imperative to strike a balance between utilizing AI tools as aids and preserving the integrity of human learning experiences (Gordijn and Have, 2023). Overreliance on AI tools can inadvertently diminish students' critical thinking and problem-solving skills, as well as hinder their ability to learn from failure and setbacks.

As Industry 4.0 unfolds with its profound technological advancements, academic institutions must rise to the occasion by reimagining curriculum design. Holistic curriculum design entails a deliberate fusion of technical proficiency with essential human skills. AI tools like ChatGPT can play a pivotal role in this transformation, acting as dynamic resources that enhance students' learning journeys by providing real-time insights, fostering interdisciplinary thinking, and facilitating innovation (Lytras, 2023). However, the curriculum's success lies in its ability to strike a balance, leveraging AI tools to amplify human potential without overshadowing the development of critical thinking, emotional intelligence, and adaptability. In designing a curriculum that reflects the needs of Industry 4.0, academic institutions can equip students not only with the technical prowess but also the holistic skill set required to thrive in the age of automation and innovation.

3.3.3. Empowering faculty through training: nurturing educators as AI guides

Over the past two decades, the significance of education, particularly within the context of teachers' education, in advancing sustainable lifestyles and SD has captured the attention of numerous scholars and educators globally (Alvarez-García et al., 2018; Kalsoom et al., 2018; Merritt et al., 2018; Tomas et al., 2015). The efficacy of educators has consistently been a topic of scholarly discourse, given that students' learning outcomes and accomplishments are heavily contingent upon the effectiveness of their teachers. According to Darling-Hammond (2017), achieving teacher effectiveness depends on well-structured professional development initiatives. Consequently, the preparatory training of prospective educators should be oriented toward proficiently integrating ESD and facilitating the holistic development of students.

Faculty empowerment and training are pivotal aspects of successfully integrating AI tools into education for Industry 4.0 and ESD (Boeve-de Pauw et al., 2022). Educators play a central role in guiding students through the complex landscape of technological advancements and sustainability challenges (Burbules et al., 2020). To fulfill this role effectively, they must themselves be proficient in AI tools and their applications. Faculty training programs should focus on enhancing educators' AI literacy, ensuring they can navigate and utilize AI tools confidently. Furthermore, educators should be empowered to become AI guides, capable of not only incorporating AI into their teaching methodologies but also guiding students on responsible and ethical AI use. Empowering faculty with the necessary knowledge and skills to harness AI for educational purposes is essential for equipping the next generation with the capabilities needed to thrive in Industry 4.0 while promoting sustainability and ethical considerations.

Pedagogical training contributes to the professional advancement of the teaching profession, amplifies student learning and advancement, and advances ESD initiatives (Faulkner et al., 2016, 2017). The education of teachers remains a pivotal element in the attainment of the United Nations' SDGs (Leal Filho et al., 2018). Emphasized by UNESCO, teacher education assumes a pivotal role in the adoption and execution

of ESD policies, which carry strategic significance in equipping forthcoming educators to instruct others in the principles of SD (Ferreira et al., 2007). Consequently, integrating ESD into pre-service teacher programs assumes a strategic essence in preparing present-day educators for the imperatives of future sustainability.

Educators assume a pivotal role in effecting the transformation of educational institutions and broader societal contexts (Ferreira et al., 2007). To catalyze this societal transformation and realize SDGs, an intricately designed and firmly established pedagogical process is imperative, fostering substantive shifts in individual cognitive frameworks (Bürgeuer and Barth, 2018). ESD serves as a meticulously devised and firmly grounded platform, fostering heightened awareness within individuals and augmenting their capacity for judicious decision-making and sustainable action (Laurie et al., 2016). Undoubtedly, ESD furnishes the requisite pedagogical apparatus for stimulating societal transformation, prompting various educational institutions to initiate endeavors spanning from school-based cultivation to certifications in sustainability, aimed at reshaping individual perspectives (Van Poeck et al., 2018). Education represents the linchpin for the realization of SD, exerting substantial influence over individual cognitive frameworks and behavioral patterns (Nousheen et al., 2020).

To maximize the impact of AI tools in education, institutions must provide ongoing and comprehensive training programs for faculty. These programs should encompass a diverse range of aspects, including the technical functionalities of AI tools (King, 2023), best practices for integration, critical evaluation of AI-generated content, and navigating ethical considerations. Faculty members should be equipped with the skills to effectively design AI-infused lessons, facilitate interactive discussions around AI-generated content, and guide students in developing critical thinking skills to evaluate responses from AI tools.

While AI tools offer efficiency and innovation, they also introduce ethical complexities and pedagogical challenges. In the realm of ethics, educators must grapple with dilemmas such as data privacy, plagiarism, and transparency, as highlighted by Else et al. (2023). These issues necessitate well-informed guidance from teachers, as they navigate the responsible use of AI-powered tools within their teaching practices. From a pedagogical perspective, faculty members should be equipped to strike a delicate balance between leveraging AI assistance and nurturing students' independent thinking and creativity, as emphasized by (Elbanna and Armstrong, 2023). This entails offering guidance on when to encourage originality and when it is appropriate to utilize AI-generated content, fostering a harmonious fusion of human and AI capabilities in the educational process.

The integration of AI tools like ChatGPT into education underscores the pivotal role of educators as guides who empower students to navigate the dynamic digital landscape. Faculty empowerment and training emerge as essential pillars to harness the full potential of AI tools while navigating ethical, pedagogical, and technological dimensions. As academic institutions embark on this transformative journey, they must invest in ongoing training initiatives that equip educators with the skills to effectively integrate AI tools, foster critical thinking, and navigate ethical considerations. By nurturing a faculty adept in leveraging AI tools, institutions lay the foundation for a future where education and technology harmoniously converge to create enriched and impactful learning experiences.

3.3.4. Rethinking Bloom's taxonomy: adapting to AI in education

In the context of integrating AI, such as ChatGPT, with Bloom's Taxonomy, it is essential to consider how AI can contribute to the educational process. The process of knowledge co-creation, emulates active involvement within the teaching-learning paradigm. It amplifies the allure of erudition, engendering heightened motivation and enthusiasm among students. They evince an augmented ardor for digital pedagogical solutions (Blau and Shamir-Inbal, 2017), with co-creation engendering a sustained dialogue between learners and educators (Catherine Bovill, 2014). This interaction not only fosters a sense of

accountability via the learning process but also cultivates the realization that their contributions are earnestly regarded by others. Consequently, students and instructional staff are inclined to esteem collaborative pedagogical planning procedures over the final learning outcomes (Catherine Bovill et al., 2011; Pandey et al., 2017). Moreover, the reservoir of students' learning enthusiasm, harnessed through this co-creation process, is directed towards acquiring essential competencies pertinent to the 21st century—competencies encompassing the ability to orchestrate personal growth and contribute to the holistic betterment of life on Earth. These competencies are cultivated and further refined on a sustained and enduring basis.

However, a central concern stemming from the use of AI-generated content, as we delve into this integration, is the potential lack of critical evaluation on the part of students (Cotton et al., 2023). As students increasingly rely on AI tools for content creation, there is a danger of uncritically accepting the information provided without subjecting it to rigorous scrutiny. Critical evaluation, a cornerstone of education, involves the skill of analyzing, questioning, and discerning the accuracy, credibility, and biases inherent in the information encountered (Deng and Lin, 2023). In an era where AI-generated content seamlessly blends with human-generated content, the ability to critically evaluate becomes paramount in distinguishing between accurate and unreliable information.

Another pressing concern arising from the integration of AI tools like ChatGPT is the potential erosion of students' authenticity and originality. The allure of instant content generation may inadvertently lead to a homogenization of ideas, where assignments and projects bear a resemblance due to their shared reliance on AI assistance (Deng and Lin, 2023). This homogenization not only stifles the diverse perspectives and unique insights that students bring to their work but also dilutes the rich tapestry of thought that should characterize educational discourse (Kleebayoon and Wiwanitkit, 2023). Moreover, the line between a student's original voice and the AI-generated content becomes increasingly blurred, raising ethical questions about the attribution of ideas.

In light of these considerations, the question emerges: should we integrate AI such as ChatGPT with Bloom's Taxonomy as a new category, explicitly addressing these AI-related challenges, or should we use AI to help achieve the existing categories, ensuring that AI serves as a tool to enhance rather than replace traditional educational objectives?

Bloom's Taxonomy (Fig. 3a), a framework for categorizing educational objectives, has been a valuable tool for educators for many years. While it provides a structured way to think about educational goals and objectives, it is not a static framework and has seen adaptations and modifications over time to meet the changing needs of education. The introduction of AI, including chatbots like ChatGPT, into education does raise some interesting questions about how we approach teaching and learning. AI can provide new tools and opportunities for personalized learning, instant feedback, and access to vast amounts of information. However, it also raises concerns about the role of human educators, the need for critical thinking skills in an age of readily available information, and the ethical use of AI in education. Modifying Bloom's Taxonomy to address the impact of AI and chatbots on education is certainly a possibility, but it should be done thoughtfully and with a clear understanding of the implications.

Integrating AI, such as ChatGPT, with Bloom's Taxonomy can be approached in multiple ways, and the choice largely depends on educational goals, contexts, and the desired outcomes. Create a new category within Bloom's Taxonomy involves adding a new category or categories specifically dedicated to AI integration (Fig. 3b). This can help explicitly emphasize the role of AI in education and ensure that AI-related skills and concepts are addressed comprehensively. This approach provides a clear framework for addressing AI-related skills and knowledge, and highlights the significance of AI in education. However, this approach may lead to a perception that AI is a separate entity from traditional educational goals and could potentially complicate the taxonomy with additional levels/categories. It may be necessary to rethink

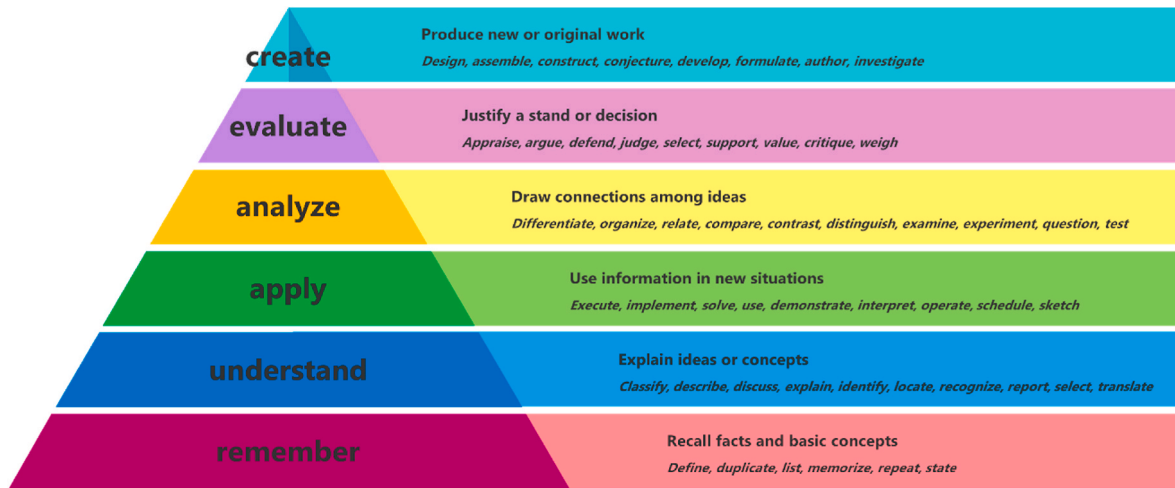


Fig. 3a. Bloom's taxonomy.



Fig. 3b. Should Bloom's Taxonomy modified due to the evolution of AI in education.?

of adding new levels or categories to Bloom's Taxonomy to account for the use of AI in education. For example, there could be a category related to "Enhance" or "Innovate." However, it is important to emphasize the responsible and ethical use of AI in education. This could be integrated into the taxonomy as a cross-cutting theme, ensuring that students are not just using AI but using it in an informed and responsible manner. Furthermore, AI can perform tasks that were previously the domain of humans, but it is important to teach students how to think critically about the information they receive from AI sources. This could be incorporated into the taxonomy's higher-order thinking skills. In addition, as AI becomes more integrated into various fields, collaboration between humans and AI will be increasingly important. This could be a focus area within the taxonomy, teaching students how to work effectively with AI systems. Finally, digital literacy skills, including the ability to navigate AI-driven technologies, could be integrated into the foundational skills of the taxonomy.

Conversely, using AI to help achieve existing categories integrates AI tools and technologies, such as ChatGPT, as tools to assist in achieving the existing categories and objectives within Bloom's Taxonomy. AI is utilized as a means to enhance traditional teaching and learning processes. This approach maintains the existing structure and familiarity of

Bloom's Taxonomy and demonstrates the practical application of AI in achieving educational goals. However, it may require educators to adapt teaching methods to incorporate AI effectively and might not explicitly address AI-related concepts and skills.

While modifying Bloom's Taxonomy to account for AI and chatbots in education is a possibility, it is essential to approach such modifications carefully and collaboratively with educators, experts, and stakeholders. The core principles of Bloom's Taxonomy—such as promoting higher-order thinking skills, active learning, and student engagement—remain relevant, but the taxonomy can evolve to address the changing educational landscape shaped by AI and technology.

4. Discussion

ESD is becoming increasingly important in the era of Industry 4.0, as it plays a critical role in preparing students for a sustainable and technologically advanced future. This involves creating an institutional ecosystem that supports sustainability principles and aligning academic curricula with the demands of the labor market and employers. It also requires the cultivation of a sustainability mindset among students, empowering them to champion sustainable transformation.

In the Industry 4.0 era, innovation and adaptability are key, and ESD promotes critical thinking, problem solving, and a holistic understanding of global challenges. By integrating sustainability principles into education, students are encouraged to think creatively and develop solutions that align with the needs of technologically advanced industries. This ensures that graduates possess the interdisciplinary skills required to contribute meaningfully to Industry 4.0 sectors. Moreover, Industry 4.0 emphasizes data-driven decision-making and advanced technologies. ESD underscores the responsible use of resources and ethical considerations, critical for the ethical and sustainable implementation of Industry 4.0 technologies. Integrating sustainability into education instills an ethical mindset among students, ensuring that they apply emerging technologies with a focus on long-term environmental and societal impacts. Industry 4.0 also thrives on collaboration and interconnectedness, and ESD often emphasizes interdisciplinary teamwork and engagement with various stakeholders. By fostering collaboration and a systems-thinking approach, HEIs prepare students to work effectively in the interconnected web of Industry 4.0.

However, aligning ESD with AI tools in Industry 4.0 presents both potentials and challenges. AI can personalize learning experiences, provide real-time data analysis, and offer data-driven insights into sustainability challenges. Nevertheless, it also raises issues like the digital divide, bias, privacy, overreliance, and resistance to change. The integration of AI tools in higher education can revolutionize sustainability education by offering relevant and up-to-date curricula, fostering cooperation between disciplines, enhancing skills relevant to Industry 4.0, and expanding opportunities for lifelong learning. Nevertheless, the challenges include addressing the digital divide, preventing AI bias, keeping up with rapid technological change, overcoming resistance to change, ensuring data privacy and security, avoiding overreliance on AI, and maintaining a holistic approach to sustainability education. HEIs must navigate these challenges to effectively harness the potential of AI for ESD in Industry 4.0. They play a pivotal role in shaping the future of education by leveraging AI's transformative potential while ensuring responsible and ethical use of AI tools.

In this evolving landscape, HEIs should adopt a comprehensive approach that includes curriculum development, pedagogical strategies, collaboration with industry partners, and the integration of emerging technologies. By doing so, they can prepare students to thrive in the era of technological transformation and contribute meaningfully to Industry 4.0 sectors. HEIs should also recognize their role in societal transformation and align their efforts with the SDGs while embracing a holistic approach to education. This involves integrating sustainability principles into campus operations, fostering community engagement, and cultivating a sustainability-conscious mindset among students.

Personalized learning is a pedagogical approach that tailors education to the unique needs, interests, and abilities of each student. With the advent of AI, personalized learning has gained significant traction as AI tools can analyze vast amounts of data to create customized learning experiences. This approach aligns well with the demands of Industry 4.0, where agility and adaptability are crucial. AI-powered systems, such as adaptive learning platforms and intelligent tutors, can provide students with content and activities that match their learning pace and style, ultimately enhancing their understanding and retention of material. AI tools like ChatGPT can play a pivotal role in personalized learning by offering real-time support and assistance to students. These tools can provide instant answers to questions, generate additional explanations, and offer targeted resources based on a student's individual learning profile. For instance, a student struggling with a particular concept can receive customized explanations and practice exercises to reinforce their understanding, while a more advanced learner can access higher-level content to challenge themselves further. This tailoring of educational content not only optimizes learning outcomes but also fosters a sense of autonomy and self-efficacy in students.

Incorporating AI into personalized learning requires the careful curation of content and the development of algorithms that can adapt to

students' changing needs. Educational institutions should collaborate with AI developers and instructional designers to create systems that align with curriculum objectives and pedagogical principles. Furthermore, AI systems should be transparent in their decision-making processes, providing both educators and students with insights into how recommendations and assessments are generated. This transparency not only builds trust but also empowers students to take ownership of their learning journeys. One of the significant advantages of AI-powered personalized learning is the ability to collect and analyze data on student performance. By continuously monitoring students' progress, AI systems can identify areas of weakness and provide timely interventions. Educators can use this data to tailor their instruction further, offering additional support to struggling students and challenging advanced learners. This data-driven approach aligns with the demands of Industry 4.0, where data analytics and insights are integral to decision-making processes.

AI tools like ChatGPT can facilitate interdisciplinary collaboration by serving as a common platform for knowledge exchange. These tools can provide instant access to information from various fields, enabling students and educators to explore topics that transcend disciplinary boundaries. AI tools can also facilitate cross-disciplinary projects, where students from different fields collaborate on real-world sustainability challenges. Educational institutions should actively promote interdisciplinary collaboration through the design of flexible curricula that allow students to explore diverse subject areas. Courses that blend elements of AI, sustainability, and other relevant disciplines can encourage students to think critically and creatively about complex global challenges. Moreover, institutions should establish interdisciplinary research centers and initiatives that bring together experts from various fields to address sustainability issues using AI and other cutting-edge technologies.

Assessing the alignment of ESD with AI tools in the context of Industry 4.0 requires a reevaluation of assessment and evaluation methods. Traditional assessment approaches often focus on measuring content knowledge through standardized tests and examinations. However, in the era of Industry 4.0, holistic competencies such as critical thinking, creativity, adaptability, and ethical reasoning are equally, if not more, important. Therefore, assessment methods should evolve to encompass these essential skills. AI tools can play a role in this transformation by offering innovative assessment techniques. For instance, AI-driven simulations and scenarios can present students with real-world challenges where they must apply their knowledge and skills to make decisions and solve problems. These simulations can provide immediate feedback, helping students understand the consequences of their actions and encouraging reflective learning. AI-powered assessments can also analyze the quality of students' written and verbal responses, evaluating not just the correctness of answers but also the depth of understanding, critical thinking, and ethical considerations. Additionally, AI can assist in the assessment of collaborative and interdisciplinary projects. It can analyze students' contributions to group work, their ability to communicate and collaborate effectively, and their capacity to synthesize knowledge from multiple disciplines. This holistic evaluation aligns with the multifaceted demands of Industry 4.0, where professionals must work collaboratively and adapt to rapidly changing circumstances.

Integrating AI, such as ChatGPT, with Bloom's Taxonomy is a complex endeavor that requires a thoughtful approach to leverage the benefits of AI while addressing the associated challenges. One approach is to create a new category within Bloom's Taxonomy dedicated to AI integration. This new category could explicitly emphasize the role of AI in education, ensuring that AI-related skills and concepts are addressed comprehensively. However, this approach may risk compartmentalizing AI as a separate entity from traditional educational goals and potentially complicate the taxonomy with additional levels or categories. Another perspective is to use AI to help achieve the existing categories within Bloom's Taxonomy. In this approach, AI is seen as a tool to enhance

traditional teaching and learning processes. This approach maintains the existing structure and familiarity of Bloom's Taxonomy while demonstrating the practical application of AI in achieving educational goals. However, it requires educators to adapt their teaching methods effectively to incorporate AI and may not explicitly address AI-related concepts and skills.

Regardless of the chosen approach, certain considerations must be taken into account when integrating AI into education. One primary concern is the potential lack of critical evaluation by students when using AI-generated content. As students increasingly rely on AI tools, they may accept information uncritically, neglecting the essential skill of critical evaluation. This skill is crucial for distinguishing between accurate and unreliable information, especially in an age where AI-generated content seamlessly blends with human-generated content. Another concern is the risk of eroding students' authenticity and originality as they become dependent on AI for content generation. Instant content generation may lead to a homogenization of ideas, where assignments and projects become similar due to their shared reliance on AI assistance. This homogenization stifles diverse perspectives and unique insights, diluting the rich tapestry of thought that should characterize educational discourse. To address these concerns, it is essential to teach students how to think critically about the information they receive from AI sources. This critical evaluation skill can be integrated into the taxonomy's higher-order thinking skills, ensuring that students are not just using AI but using it in an informed and responsible manner. Additionally, the responsible and ethical use of AI in education should be emphasized as a cross-cutting theme within the taxonomy.

5. Policy implications

In the rapidly changing landscape of education in the era of Industry 4.0, it is crucial for education policies and strategies of academic institutions to be meticulously designed to meet the dynamic needs of students and society. The research findings discussed in the previous sections offer valuable insights that can guide the formulation of these policies and strategies, paving the way for a more effective and responsive education system. Some of these policies and strategies include:

Education policies should encourage academic institutions to adopt flexible and adaptable curricula. These curricula should blend technical knowledge with higher-order cognitive skills to meet the demands of Industry 4.0. Strategies might involve periodic curriculum reviews, interdisciplinary courses, and experiential learning opportunities that bridge the gap between theory and real-world application. This approach ensures that students receive a well-rounded education that prepares them for the complexities of the modern job market.

A key implication is the need to prioritize comprehensive training programs for educators. These programs should ensure that educators are well-versed in AI technologies, ethical considerations, and innovative teaching methods. Strategies could include workshops, seminars, and professional development opportunities that empower educators to effectively integrate AI tools into their teaching methodologies. This training equips them with the skills necessary to encourage critical thinking among students and navigate the ethical challenges associated with AI in education.

Another crucial implication is the integration of ethical AI education throughout the curriculum. Policies should emphasize the creation of dedicated AI ethics courses, workshops, and awareness campaigns. These initiatives ensure that students are well-informed about the ethical implications of AI tools and their responsible use. By nurturing an understanding of AI ethics, academic institutions can produce graduates who are not only technically proficient but also ethically conscious.

Education policies must promote a culture of continuous learning that equips students with the skills needed to thrive in an ever-changing landscape. Strategies might include offering micro-credentialing

options, supporting self-directed learning through digital platforms, and fostering a growth mindset among students. In this way, students are prepared not just for their initial careers but also for lifelong learning in the face of evolving technologies.

Policies should encourage academic-industry partnership and collaborations to ensure graduates are job-ready and aligned with industry needs. Strategies could include incentivizing partnerships, establishing advisory boards with industry experts, and facilitating joint research projects that bridge the gap between academia and the workforce. By fostering strong ties with industries, academic institutions can offer students practical insights and experiences, enhancing their employability.

Prioritizing inclusivity and diversity in education is essential. Policies may involve implementing measures to bridge the digital divide, addressing biases in AI algorithms, and creating a diverse and inclusive learning environment that reflects the realities of a globalized and interconnected world. In doing so, academic institutions can ensure that education is accessible to all and that diverse perspectives enrich the learning experience.

Policies should consider innovative assessment methods that evaluate not only technical knowledge but also critical thinking, problem-solving abilities, and creativity. Strategies could involve incorporating project-based assessments, peer evaluations, and real-world case studies to measure students' holistic skill development. These assessments provide a more accurate reflection of students' abilities and encourage the development of essential skills.

Policies must allocate resources to support the implementation of new educational technologies, faculty development programs, and infrastructure improvements. Strategies could involve securing funding for technology integration, modernizing classroom facilities, and ensuring access to online learning resources. Adequate resource allocation is vital to the successful implementation of educational policies.

Policies should encourage collaboration and communication among various stakeholders, including educators, students, industry partners, policymakers, and the wider community. Strategies might involve establishing forums for dialogue, conducting surveys to gather feedback, and incorporating diverse perspectives in policy formulation. Engaging stakeholders ensures that policies are responsive to the needs and expectations of all involved parties.

Lastly, policies should consider the ethical and legal implications of AI integration in education. Strategies could involve developing guidelines for data privacy, intellectual property rights, and responsible AI use within educational contexts. A well-defined regulatory framework safeguards against potential risks and ensures the ethical use of AI in education.

6. Conclusion

This study has laid the foundation for a thorough examination of the integration of AI tools in education within the context of ESD during the Industry 4.0 era. We have witnessed the profound transformation brought about by Industry 4.0, characterized by a multitude of advanced technologies that promise numerous advantages for businesses and industries. However, unlocking the full potential of Industry 4.0 hinges on addressing the need for individualized proficiencies and expertise within the workforce - a challenge that HEIs are well-positioned to tackle.

The importance of ESD, underscored by global initiatives and the imperative for sustainability, is intricately tied to the evolving landscape of education. The integration of AI tools into education has emerged as a transformative catalyst, presenting both promises and complexities. AI, epitomized by advanced language models like ChatGPT, holds the potential to revolutionize pedagogy, elevate personalized learning, and stimulate innovation. Yet, as these tools become more prevalent in educational settings, ethical considerations, curriculum design, continuous learning, and alignment with industry demands come to the

forefront. Additionally, the study has shed light on substantial gaps in the current body of literature, emphasizing the imperative for a systematic and rigorous exploration of AI's influence.

This paper underscores the crucial role of ESD in the era of Industry 4.0, emphasizing its significance in preparing students for a sustainable and technologically advanced future. The discussion has illuminated the potential benefits and significant challenges of integrating AI tools into ESD. It has become evident that ESD equips students with essential skills such as critical thinking, problem-solving, and interdisciplinary knowledge, aligning perfectly with the demands of Industry 4.0. On the other hand, AI tools offer personalized learning experiences, foster interdisciplinary collaboration, and enhance student autonomy, providing a promising avenue for education. However, this integration also poses challenges, including addressing the digital divide, AI bias, data privacy, and the need to instill responsible AI use. To effectively harness the potential of AI in education, comprehensive policies and strategies are necessary, encompassing flexible curricula, faculty development, ethical AI integration, a culture of continuous learning, industry partnerships, inclusivity, innovative assessment methods, adequate resource allocation, and stakeholder engagement. Additionally, a well-defined regulatory framework is essential to ensure the ethical use of AI in education. Ultimately, this synthesis presents a transformative opportunity to shape a generation of learners equipped not only with technical skills but also with critical thinking, ethical awareness, and adaptability to thrive in the era of Industry 4.0. Collaboration among educational institutions, policymakers, educators, and students is imperative to realize this vision and pave the way for a sustainable and technologically advanced future.

CRediT authorship contribution statement

Ammar Abulibdeh: Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Esmat Zaidan:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Rawan Abulibdeh:** Conceptualization, Data curation, Formal analysis, Methodology, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- Agathokleous, E., Saitanis, C.J., Fang, C., Yu, Z., 2023. Use of ChatGPT: what does it mean for biology and environmental science? *Sci. Total Environ.* 888, 164154. <https://doi.org/10.1016/J.SCITOTENV.2023.164154>.
- Ahmad, S.F., Alam, M.M., Rahmat, M.K., Shahid, M.K., Aslam, M., Salim, N.A., Al-Abyadh, M.H.A., 2023. Leading edge or bleeding edge: Designing a framework for the adoption of AI technology in an educational organization. *Sustainability* 15 (8), 6540. <https://doi.org/10.3390/SU15086540>.
- AlMalki, H.A., Durugbo, C.M., 2023. Evaluating critical institutional factors of Industry 4.0 for education reform. *Technol. Forecast. Soc. Change* 188, 122327. <https://doi.org/10.1016/J.TECHFORE.2023.122327>.
- Alvarez-García, O., Sureda-Negre, J., Comas-Forgas, R., 2018. Assessing environmental competencies of primary education pre-service teachers in Spain: a comparative study between two universities. *Int. J. Sustain. High Educ.* 19 (1), 15–31. <https://doi.org/10.1108/IJSHE-12-2016-0227/FULL/XML>.
- Amaral, A., Peças, P., 2021. SMEs and Industry 4.0: two case studies of digitalization for a smoother integration. *Comput. Ind.* 125, 103333. <https://doi.org/10.1016/J.COMPIND.2020.103333>.
- Ambusaidi, A., Al Washahi, M., 2016. Prospective teachers' perceptions about the concept of sustainable development and related issues in Oman. *J. Educ. Sustain. Dev.* 10 (1), 3–19. <https://doi.org/10.1177/0973408215625528>.
- Annan-Diab, F., Molinari, C., 2017. Interdisciplinarity: practical approach to advancing education for sustainability and for the sustainable development goals. *Int. J. Manag. Educ.* 15 (2), 73–83. <https://doi.org/10.1016/J.IJME.2017.03.006>.
- Aoun, A., Ilinca, A., Ghandour, M., Ibrahim, H., 2021. A review of Industry 4.0 characteristics and challenges, with potential improvements using blockchain technology. *Comput. Ind. Eng.* 162, 107746. <https://doi.org/10.1016/J.CIE.2021.107746>.
- Argento, D., Einarson, D., Mårtensson, L., Persson, C., Wendin, K., Westergren, A., 2020. Integrating sustainability in higher education: a Swedish case. *Int. J. Sustain. High Educ.* 21 (6), 1131–1150. <https://doi.org/10.1108/IJSHE-10-2019-0292/FULL/XML>.
- Baena-Morales, S., Fröberg, A., 2023. Towards a more sustainable future: simple recommendations to integrate planetary health into education. *Lancet Planet. Health* 7 (10), e868–e873. [https://doi.org/10.1016/S2542-5196\(23\)00194-8](https://doi.org/10.1016/S2542-5196(23)00194-8).
- Baidoo-Anu, D., Owusu Ansah, L., 2023. Education in the era of generative artificial intelligence (AI): Understanding the potential benefits of ChatGPT in promoting teaching and learning. Available at SSRN 4337484, 2023. [papers.Ssrn.Com. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4337484](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4337484).
- Baker-Shelley, A., van Zeijl-Rozema, A., Martens, P., 2017. A conceptual synthesis of organisational transformation: how to diagnose, and navigate, pathways for sustainability at universities? *J. Clean. Prod.* 145, 262–276. <https://doi.org/10.1016/J.JCLEPRO.2017.01.026>.
- Beynaghi, A., Trencher, G., Moztarzadeh, F., Mozafari, M., Maknoon, R., Leal Filho, W., 2016. Future sustainability scenarios for universities: moving beyond the United Nations decade of education for sustainable development. *J. Clean. Prod.* 112, 3464–3478. <https://doi.org/10.1016/J.JCLEPRO.2015.10.117>.
- Bibri, S.E., Krogstie, J., Kaboli, A., Alahi, A., 2024. Smarter eco-cities and their leading-edge artificial intelligence of things solutions for environmental sustainability: a comprehensive systematic review. *Environ. Sci. Ecotechnol.* 19, 100330. <https://doi.org/10.1016/J.ESE.2023.100330>.
- Bildirici, M., Ersin, Ö., 2023. Nexus between Industry 4.0 and environmental sustainability: a Fourier panel bootstrap cointegration and causality analysis. *J. Clean. Prod.* 386, 135786. <https://doi.org/10.1016/J.JCLEPRO.2022.135786>.
- Biswas, S.S., 2023. Role of Chat GPT in public health. *Ann. Biomed. Eng.* 51 (5), 868–869. <https://doi.org/10.1007/S10439-023-03172-7/METRICS>.
- Blau, I., Shamir-Inbal, T., 2017. Re-designed flipped learning model in an academic course: the role of co-creation and co-regulation. *Comput. Educ.* 115, 69–81. <https://doi.org/10.1016/J.COMPEDU.2017.07.014>.
- Boeve-de Pauw, J., Olsson, D., Berglund, T., Gericke, N., 2022. Teachers' ESD self-efficacy and practices: a longitudinal study on the impact of teacher professional development. *Environ. Educ. Res.* 28 (6), 867–885. <https://doi.org/10.1080/13504622.2022.2042206>.
- Bonfield, C.A., Salter, M., Longmuir, A., Benson, M., Adachi, C., 2020. Transformation or evolution?: education 4.0, teaching and learning in the digital age. *Higher Education Pedagogies* 5 (1), 223–246. <https://doi.org/10.1080/23752696.2020.1816847>.
- Bovill, C., Cook-Sather, A., Felten, P., Millard, L., Moore-Cherry, N., 2016. Addressing potential challenges in co-creating learning and teaching: overcoming resistance, navigating institutional norms and ensuring inclusivity in student-staff partnerships. *High Educ.* 71 (2), 195–208. <https://doi.org/10.1007/S10734-015-9896-4/FIGURES/1>.
- Bovill, Catherine, 2014. An investigation of co-created curricula within higher education in the UK, Ireland and the USA. *Innovat. Educ. Teach. Int.* 51 (1), 15–25. <https://doi.org/10.1080/14703297.2013.770264>.
- Bovill, Catherine, Cook-Sather, A., Felten, P., 2011. Students as co-creators of teaching approaches, course design, and curricula: implications for academic developers. *Int. J. Acad. Dev.* 16 (2), 133–145. <https://doi.org/10.1080/1360144X.2011.568690>.
- Brahma, M., Tripathi, S.S., Sahay, A., 2021. Developing curriculum for industry 4.0: digital workplaces. *High Educ. Skills Work. base Learn.* 11 (1), 144–163. <https://doi.org/10.1108/HESWBL-08-2019-0103/FULL/XML>.
- Buçinca, Z., Malaya, M.B., Gajos, K.Z., 2021. To trust or to think. *Proceed. ACM on Human-Comput. Inter.* 5 (CSCW1) <https://doi.org/10.1145/3449287>.
- Burbules, N.C., Fan, G., Repp, P., 2020. Five trends of education and technology in a sustainable future. *Geograp. Sustain.* 1 (2), 93–97. <https://doi.org/10.1016/J.GEOSUS.2020.05.001>.
- Bürgener, L., Barth, M., 2018. Sustainability competencies in teacher education: Making teacher education count in everyday school practice. *J. Clean. Prod.* 174, 821–826. <https://doi.org/10.1016/J.JCLEPRO.2017.10.263>.
- Burnard, P., 2006. Reflecting on the creativity agenda in education, 36 (3), 313–318. <https://doi.org/10.1080/03057640600865801>.
- Caiado, R.G.G., Scavarda, L.F., Gavião, L.O., Ivson, P., Nascimento, D.L. de M., Garza-Reyes, J.A., 2021. A fuzzy rule-based industry 4.0 maturity model for operations and supply chain management. *Int. J. Prod. Econ.* 231, 107883. <https://doi.org/10.1016/J.IJPE.2020.107883>.
- Casarejos, F., Gustavson, L.M., Frota, M.N., 2017. Higher Education Institutions in the United States: commitment and coherency to sustainability vis-à-vis dimensions of the institutional environment. *J. Clean. Prod.* 159, 74–84. <https://doi.org/10.1016/J.JCLEPRO.2017.05.034>.
- Cebrián, G., Palau, R., Mogas, J., 2020. The smart classroom as a means to the development of ESD methodologies. *Sustainability* 12 (7), 3010. <https://doi.org/10.3390/SU12073010>.
- Cebrián, G., Junyent, M., Mulà, I., 2021. Current practices and future pathways towards competencies in education for sustainable development. *Sustainability* 13 (16), 8733. <https://doi.org/10.3390/SU13168733>.

- Ching, N.T., Ghobakhloo, M., Iranmanesh, M., Maroufkhani, P., Asadi, S., 2022. Industry 4.0 applications for sustainable manufacturing: a systematic literature review and a roadmap to sustainable development. *J. Clean. Prod.* 334, 130133 <https://doi.org/10.1016/j.jclepro.2021.130133>.
- Cortese, A.D., 2003. The critical role of higher education in creating a sustainable future. *Plann. High. Educ.* 31, 15–22.
- Cotton, D.R.E., Cotton, P.A., Shipway, J.R., 2023. Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. <https://doi.org/10.1080/14703297.2023.2190148>.
- Day, T., 2023. A Preliminary Investigation of Fake Peer-Reviewed Citations and References Generated by ChatGPT. *Professional Geographer*. <https://doi.org/10.1080/00330124.2023.2190373>.
- Del Río Castro, G., González Fernández, M.C., Uruburu Colsa, Á., 2021. Unleashing the convergence amid digitalization and sustainability towards pursuing the Sustainable Development Goals (SDGs): a holistic review. *J. Clean. Prod.* 280, 122204 <https://doi.org/10.1016/j.jclepro.2020.122204>.
- Deng, J., Lin, Y., 2023. The benefits and challenges of ChatGPT: an overview. *Frontiers in Comput. Intelligent Systems* 2 (2), 81–83. <https://doi.org/10.54097/fcis.v2i2.4465>.
- Dilmaghani, S., Brust, M.R., Danoy, G., Cassagnes, N., Pecero, J., Bouvry, P., 2019. Privacy and Security of Big Data in AI Systems: A Research and Standards Perspective, pp. 5737–5743. *Proceedings - 2019 IEEE International Conference on Big Data* 2019. <https://doi.org/10.1109/BIGDATA47090.2019.9006283>.
- Dinca, C., Rousseaux, P., Badea, A., 2007. A life cycle impact of the natural gas used in the energy sector in Romania. *J. Clean. Prod.* 15 (15), 1451–1462. <https://doi.org/10.1016/j.jclepro.2006.03.011>.
- van Dis, Bollen, J., Zuidema, W., van Rooij, R., Bockting, C.L., 2023. ChatGPT: five priorities for research. *Nature* 614 (7947), 224–226. <https://doi.org/10.1038/d41586-023-00288-7>.
- Disterheft, A., Caeiro, S., Azeiteiro, U.M., Filho, W.L., 2015. Sustainable universities – a study of critical success factors for participatory approaches. *J. Clean. Prod.* 106, 11–21. <https://doi.org/10.1016/j.jclepro.2014.01.030>.
- Dlouhá, J., Glavič, P., Barton, A., 2017. Higher education in Central European countries – critical factors for sustainability transition. *J. Clean. Prod.* 151, 670–684. <https://doi.org/10.1016/j.jclepro.2016.08.022>.
- Dwivedi, Y.K., Hughes, L., Ismagilova, E., Aarts, G., Coombs, C., Crick, T., Duan, Y., Dwivedi, R., Edwards, J., Eirug, A., Galanos, V., Ilavarasan, P.V., Janssen, M., Jones, P., Kar, A.K., Kizgin, H., Kronemann, B., Lal, B., Lucini, B., Williams, M.D., 2021. Artificial Intelligence (AI): Multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. *Int. J. Inform. Manag.* 57, 101994. <https://doi.org/10.1016/j.ijinfomgt.2019.08.002>.
- Elbanna, S., Armstrong, L., 2023. Exploring the integration of ChatGPT in education: adapting for the future. *Manag. Sustain. Arab Rev.* (ahead-of-print) <https://doi.org/10.1108/MSAR-03-2023-0016>.
- Else, H., Else, Holly, 2023. Abstracts written by ChatGPT fool scientists. *Nature* 613 (7944). <https://doi.org/10.1038/d41586-023-00056-7>.
- Elton, L., 2003. Dissemination of innovations in higher education: a change theory approach. *Tert. Educ. Manag.* 9 (3), 199–214. <https://doi.org/10.1023/A:1024472813449>.
- Faulkner, S.A., Cook, C.M., Thompson, N.L., Howell, P.B., Rintamaa, M.F., Miller, N.C., 2017. Mapping the varied terrain of specialized middle level teacher preparation and licensure. *Middle Sch. J.* 48 (2), 8–13. <https://doi.org/10.1080/00940771.2017.1272911>.
- Fekih Zguir, M., Dubis, S., Koc, M., 2021. Embedding Education for Sustainable Development (ESD) and SDGs values in curriculum: a comparative review on Qatar, Singapore and New Zealand. *J. Clean. Prod.* 319, 128534 <https://doi.org/10.1016/j.jclepro.2021.128534>.
- Ferreira, J., Ryan, L., Tilbury, D., 2007. Mainstreaming education for sustainable development in initial teacher education in Australia: A review of existing professional development models. *J. Edu. Teach.* 33 (2), 225–239. Taylor & Francis. <https://doi.org/10.1080/02607470701259515>.
- Fuchs, K., 2023. Exploring the opportunities and challenges of NLP models in higher education: is Chat GPT a blessing or a curse? *Frontiers in Educ.* 8, 1166682 <https://doi.org/10.3389/FEDUC.2023.1166682/BIBTEX>.
- Fuertes, J.J., Prada, M.A., Rodriguez-Ossorio, J.R., Gonzalez-Herbon, R., Perez, D., Dominguez, M., 2021. Environment for Education on Industry 4.0. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2021.3120517>.
- García-González, E., Jiménez-Fontana, R., Goded, P.A., 2020. Approaches to teaching and learning for sustainability: Characterizing students' perceptions. *J. Clean. Prod.* 274, 122928 <https://doi.org/10.1016/j.jclepro.2020.122928>.
- Gaube, S., Suresh, H., Raue, M., Merritt, A., Berkowitz, S.J., Lerner, E., Coughlin, J.F., Guttig, J.V., Colak, E., Ghassemi, M., 2021. Do as AI say: susceptibility in deployment of clinical decision-aids. *Npj Digital Med.* 4 (1), 1–8. <https://doi.org/10.1038/s41746-021-00385-9>.
- Geithner, S., Menzel, D., 2016. Effectiveness of learning through experience and reflection in a project management simulation, 47 (2), 228–256. <https://doi.org/10.1177/1046878115624312>.
- Gill, S., Singh, G., 2020. Developing inclusive and quality learning environments in HEIs. *Int. J. Educ. Manag.* 34 (5), 823–836. <https://doi.org/10.1108/IJEM-03-2019-0106/FULL/XML>.
- Gordijn, B., Have, H. ten, 2023. ChatGPT: evolution or revolution? *Med. Healthc. Philos.* 26 (1), 1–2. <https://doi.org/10.1007/s11019-023-10136-0/METRCS>.
- Grimus, M., 2020. Emerging Technologies: Impacting Learning, Pedagogy and Curriculum Development, pp. 127–151. https://doi.org/10.1007/978-981-15-0618-5_8.
- Gupta, H., Kumar, A., Wasan, P., 2021. Industry 4.0, cleaner production and circular economy: an integrative framework for evaluating ethical and sustainable business performance of manufacturing organizations. *J. Clean. Prod.* 295, 126253 <https://doi.org/10.1016/j.jclepro.2021.126253>.
- Hammad Mian, S., Salah, B., Ameen, W., Moiduddin, K., Alkhalefah, H., 2020. Adapting universities for sustainability education in industry 4.0: Channel of challenges and opportunities. *Mdpi.ComSH Mian, B Salah, W Ameen, K Moiduddin, H AlkhalefahSustainability*, 2020•mdpi.Com 12, 6100. <https://doi.org/10.3390/su12156100>.
- Han, S., 2020. Reproducing the working class? Incongruence between the valuation of social-emotional skills in school and in the labor market, 64 (3), 467–487. <https://doi.org/10.1177/0731121420956378>.
- Hensley, N., 2020. Educating for sustainable development: cultivating creativity through mindfulness. *J. Clean. Prod.* 243, 118542 <https://doi.org/10.1016/j.jclepro.2019.118542>.
- Hugé, J., Block, T., Waas, T., Wright, T., Dahdouh-Guebas, F., 2016. How to walk the talk? Developing actions for sustainability in academic research. *J. Clean. Prod.* 137, 83–92. <https://doi.org/10.1016/j.jclepro.2016.07.010>.
- Ihsan, M., Ahmad, S., Ikram Idrus, M., Rijal, S., 2023. The role of education in fostering entrepreneurial spirit in the young generation. *J. Contemp. Administr. Manag. (ADMAN)* 1 (2), 93–100. <https://doi.org/10.61100/ADMAN.V1I2.28>.
- Jan, Z., Ahamed, F., Mayer, W., Patel, N., Grossmann, G., Stumptner, M., Kuusk, A., 2023. Artificial intelligence for industry 4.0: systematic review of applications, challenges, and opportunities. *Expert Syst. Appl.* 216, 119456 <https://doi.org/10.1016/j.eswa.2022.119456>.
- Jora, O.-D., Apăvăloaei, M.-A., Stamate-Ştefan, A., Roşca, V.I., Gherghina, R., 2020. “Citadel of Science” and “Cathedral of Conscience”: The University, as institution, was born to be concerned with sustainable. *Ceeol. Com* 22 (54), 313. <https://doi.org/10.24818/EA/2020/54/312>.
- Kalsoom, Q., Khanam, A., 2017. Inquiry into sustainability issues by preservice teachers: a pedagogy to enhance sustainability consciousness. *J. Clean. Prod.* 164, 1301–1311. <https://doi.org/10.1016/j.jclepro.2017.07.047>.
- Kalsoom, Q., Qureshi, N., Khanam, A., 2018. Perceptions of the research scholars regarding education for sustainable development (ESD) in Pakistan. *World Sustain. Series* 165–179. https://doi.org/10.1007/978-3-319-73293-0_10/COVER.
- Kar, A.K., Choudhary, S.K., Singh, V.K., 2022. How can artificial intelligence impact sustainability: a systematic literature review. *J. Clean. Prod.* 376, 134120 <https://doi.org/10.1016/j.jclepro.2022.134120>.
- Kemper, J.A., Ballantine, P.W., Hall, C.M., 2020. The role that marketing academics play in advancing sustainability education and research. *J. Clean. Prod.* 248, 119229 <https://doi.org/10.1016/j.jclepro.2019.119229>.
- Khan, I.S., Ahmad, M.O., Majava, J., 2023. Industry 4.0 innovations and their implications: an evaluation from sustainable development perspective. *J. Clean. Prod.* 405, 137006 <https://doi.org/10.1016/j.jclepro.2023.137006>.
- King, M.R., 2023. A conversation on artificial intelligence, chatbots, and plagiarism in higher education. *Cell. Mol. Bioeng.* 16 (1), 1–2. <https://doi.org/10.1007/S12195-022-00754-8/FIGURES/1>.
- Kleebayon, A., Wivanitkit, V., 2023. Artificial intelligence, chatbots, plagiarism and basic Honesty: Comment. *Cell. Mol. Bioeng.* 16 (2), 173–174. <https://doi.org/10.1007/S12195-023-00759-X/METRCS>.
- Klimova, A., Rondeau, E., Andersson, K., Porras, J., Rybin, A., Zaslavsky, A., 2016. An international Master's program in green ICT as a contribution to sustainable development. *J. Clean. Prod.* 135, 223–239. <https://doi.org/10.1016/j.jclepro.2016.06.032>.
- Lambrechts, W., Mulà, I., Ceulemans, K., Molderez, I., Gaeremynck, V., 2013. The integration of competences for sustainable development in higher education: an analysis of bachelor programs in management. *J. Clean. Prod.* 48, 65–73. <https://doi.org/10.1016/j.jclepro.2011.12.034>.
- Laurie, R., Nonoyama-Tarumi, Y., Mckeown, R., Hopkins, C., 2016. Contributions of education for sustainable development (ESD) to quality education: a synthesis of research. *J. Educ. Sustain. Dev.* 10 (2), 226–242. <https://doi.org/10.1177/0973408216661442>.
- Leal Filho, W., Raath, S., Lazzarini, B., Vargas, V.R., de Souza, L., Anholon, R., Quelhas, O.L.G., Haddad, R., Klavins, M., Orlovic, V.L., 2018. The role of transformation in learning and education for sustainability. *J. Clean. Prod.* 199, 286–295. <https://doi.org/10.1016/j.jclepro.2018.07.017>.
- Leal Filho, Frankenberger, F., Salvia, A.L., Azeiteiro, U., Alves, F., Castro, P., Will, M., Platje, J., Lovren, V.O., Brandli, L., Price, E., Doni, F., Mifsud, M., Ávila, L.V., 2021. A framework for the implementation of the Sustainable Development Goals in university programmes. *J. Clean. Prod.* 299, 126915 <https://doi.org/10.1016/j.jclepro.2021.126915>.
- Leow, A., Billett, S., Le, A.H., 2023. Towards a continuing education and training eco system: a case study of Singapore. <https://doi.org/10.1080/14480220.2023.2203944>.
- Lozano, R., 2006. Incorporation and institutionalization of SD into universities: breaking through barriers to change. *J. Clean. Prod.* 14 (9–11), 787–796. <https://doi.org/10.1016/j.jclepro.2005.12.010>.
- Li, X., Zhang, T., 2017. An exploration on artificial intelligence application: From security, privacy and ethic perspective, pp. 416–420, 2017 2nd IEEE International Conference on Cloud Computing and Big Data Analysis, ICCCBDA 2017. <https://doi.org/10.1109/ICCCBDA.2017.7951949>.
- Lozano, R., Lozano, F.J., Mulder, K., Huisingh, D., Waas, T., 2013. Advancing higher education for sustainable development: international insights and critical reflections. *J. Clean. Prod.* 48, 3–9. <https://doi.org/10.1016/j.jclepro.2013.03.034>.
- Lukman, R., Glavič, Peter, 2007. What are the key elements of a sustainable university? *Clean Technol. Environ. Pol.* 9 (2), 103–114. Springer. <https://doi.org/10.1007/s10098-006-0070-7>.

- Lytras, M.D., 2023. Active and transformative learning (ATL) in higher education in times of artificial intelligence and ChatGPT: Investigating a new value-based framework. *Active and Transformative Learning in STEAM Disciplines: From Curriculum Design to Social Impact* 5–23. <https://doi.org/10.1108/978-1-83753-618-420231001/FULL/XML>.
- Mampane, S.T., 2020. Futuristic Learning: Improving Classroom Engagement and Learning in a Distance Higher Education Institution, vol. 27, pp. 189–204. <https://doi.org/10.1108/S2055-36412020000027015>.
- Merritt, E., Hale, A., Archambault, L., 2018. Changes in pre-service teachers' values, sense of agency, motivation and consumption practices: A case study of an education for sustainability course. *Sustainability* 11 (1), 155. <https://doi.org/10.3390/S11010155>.
- Mhlanga, D., 2023. Open AI in education, the responsible and ethical use of ChatGPT towards lifelong learning. *SSRN Electron. J.* <https://doi.org/10.2139/SSRN.4354422>.
- Mogali, S.R., 2023. Initial Impressions of ChatGPT for Anatomy Education. *Anatomical Sciences Education*. <https://doi.org/10.1002/ASE.2261>.
- Mourtzis, D., Angelopoulos, J., Panopoulos, N., 2022. A literature review of the challenges and opportunities of the transition from industry 4.0 to society 5.0. *Energies* 15 (17), 6276. <https://doi.org/10.3390/EN15176276>.
- Mulá, I., Tilbury, D., Ryan, A., Mader, M., Dlouhá, J., Mader, C., Benayas, J., Dlouhý, J., Alba, D., 2017. Catalysing Change in Higher Education for Sustainable Development: a review of professional development initiatives for university educators. *Int. J. Sustain. High Educ.* 18 (5), 798–820. <https://doi.org/10.1108/IJSHE-03-2017-0043/FULL/HTML>.
- Mulder, K.F., 2010. Don't preach. Practice! Value laden statements in academic sustainability education. *Int. J. Sustain. High Educ.* 11 (1), 74–85. <https://doi.org/10.1108/14676371011010066/FULL/XML>.
- Musariwa, P., Tinonetsana, F., 2023. An assessment of university in entrepreneurship training as a means of reducing youth unemployment in South Africa: a case of durban university of technology. *African J. Inter/Multidisciplin. Studies* 5 (1), 1–10. <https://doi.org/10.51415/AJIMS.V5I1.1248>.
- Neamțu, D.M., Bejinaru, R., Hapenciuc, C.V., Condratov, I., Stanciu, P., 2020. Analysis and modelling of influence factors in the configuration of a sustainable university. Case study: "Teofil cel mare. Ceeol.ComDM Neamțu, R Bejinaru, CV Hapenciuc, I Condratov, P StanciuAmfiteatru Economic, 2020•ceool.Com 22 (54), 391. <https://doi.org/10.24818/EA/2020/54/391>.
- Nousheen, A., Yousuf Zai, S.A., Waseem, M., Khan, S.A., 2020. Education for sustainable development (ESD): effects of sustainability education on pre-service teachers' attitude towards sustainable development (SD). *J. Clean. Prod.* 250, 119537. <https://doi.org/10.1016/J.JCLEPRO.2019.119537>.
- Paletta, A., Fava, F., Ubertini, F., Bastioli, C., Gregori, G., Camera, F. La, Douvan, A.R., 2019. Universities, industries and sustainable development: outcomes of the 2017 G7 environment ministerial meeting. *Sustain. Prod. Consum.* 19, 1–10. <https://doi.org/10.1016/J.SPC.2019.02.008>.
- Pandey, V., Amir, A., Debelius, J., Hyde, E.R., Kosciolk, T., Knight, R., Klemmer, S., 2017. Gut instinct: Creating scientific theories with online learners. In: *Conference on Human Factors in Computing Systems - Proceedings, 2017-May*, pp. 6825–6836. <https://doi.org/10.1145/3025453.3025769>.
- Pascoal-Faria, P., da Silva, D.P., Mateus, A., Mitchell, G.R., 2023. Digitalisation of material science – improving product design in the context of Industry 4.0. *Mater. Today: Proc.* <https://doi.org/10.1016/J.MATPR.2023.06.237>.
- Peters, M.D.J., Godfrey, C.M., Khalil, H., McInerney, P., Parker, D., Soares, C.B., 2015. Guidance for conducting systematic scoping reviews. *Int. J. Evid. Base. Healthc.* 13 (3), 141–146. <https://doi.org/10.1097/XEB.0000000000000050>.
- Pham, M.T., Rajić, A., Greig, J.D., Sargeant, J.M., Papadopoulos, A., McEwen, S.A., 2014. A scoping review of scoping reviews: advancing the approach and enhancing the consistency. *Res. Synth. Methods* 5 (4), 371–385. <https://doi.org/10.1002/JRSM.1123>.
- Pigozzi, M.J., 2007. Quality in Education Defines ESD, 1 (1), 27–35. <https://doi.org/10.1177/097340820700100108>.
- Piroșcă, G., Șerban-Oprescu, G., Badea, L., Econ, S. D.-A., 2020. Education for sustainable development—an evaluation of students' perceptions within the Bucharest University of Economic Studies. *Ceeol.ComGI Piroșcă, GL Șerban-Oprescu, L Badea, S DeduAmfiteatru Econ, 2020 •ceool.Com 22 (54). <https://doi.org/10.24818/EA/2020/54/346>.*
- Plaiaș, I., Pop, C., Băbuț, R., Dabija, DC, 2011. Employers' perception of competences acquired through academic marketing training in knowledge based economy. *Amfiteatru Econ.* 13 (30). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2305238.
- Pocol, C.B., Stanca, L., Dabija, D.C., Pop, I.D., Mișcoiu, S., 2022. Knowledge Co-creation and sustainable education in the labor market-driven university-business environment. *Front. Environ. Sci.* 10 <https://doi.org/10.3389/FENVS.2022.781075/FULL>.
- Rodríguez-Amigo, A., Fernández-Alvarado, J.F., Fernández-Rodríguez, S., 2022. Case of study on a sustainability building: environmental risk assessment related with allergenicity from air quality considering meteorological and urban green infrastructure data on BIM. *Sci. Total Environ.* 838, 155910. <https://doi.org/10.1016/J.SCITOTENV.2022.155910>.
- Romani-Dias, M., Carneiro, J., Barbosa, A., dos, S., 2019. Internationalization of higher education institutions: the underestimated role of faculty. *Int. J. Educ. Manag.* 33 (2), 300–316. <https://doi.org/10.1108/IJEM-07-2017-0184/FULL/XML>.
- Rudolph, J., Tan, S., 2023. ChatGPT: Bullshit spewer or the end of traditional assessments in higher education? *J. Appl. Learn. Teach.* 6 (1). <https://doi.org/10.37074/jalt.2023.6.1.9>.
- Sammalisto, K., Sundström, A., Holm, T., 2015. Implementation of sustainability in universities as perceived by faculty and staff – a model from a Swedish university. *J. Clean. Prod.* 106, 45–54. <https://doi.org/10.1016/J.JCLEPRO.2014.10.015>.
- Sánchez-Carracedo, F., Sureda, B., Moreno-Pino, F.M., Romero-Portillo, D., 2021. Education for Sustainable Development in Spanish engineering degrees. Case study. *J. Clean. Prod.* 294, 126322. <https://doi.org/10.1016/J.JCLEPRO.2021.126322>.
- Sanina, A., Kuterina, E., Balashov, A., 2020. The Co-Creative approach to digital simulation games in social science education. *Comput. Educ.* 149, 103813. <https://doi.org/10.1016/J.COMPEDU.2020.103813>.
- Schiele, H., Bos-Nehles, A., Delke, V., Stegmaier, P., Torn, R.J., 2022. Interpreting the industry 4.0 future: technology, business, society and people. *J. Bus. Strat.* 43 (3), 157–167. <https://doi.org/10.1108/JBS-08-2020-0181/FULL/XML>.
- Șerban, E., Hristea, A., Curea, S., Econ, R. C.-A., 2020. Sustainable Universities, from indifference to joint action—A panel data analysis. *Ceeol.Com.* <https://doi.org/10.24818/EA/2020/54/376>.
- Shahjahan, R.A., Estera, A.L., Surla, K.L., Edwards, K.T., 2022. "Decolonizing" curriculum and pedagogy: A comparative review across disciplines and global higher education contexts. *Rev. Educ. Res.* 92 (1), 73–113. <https://doi.org/10.3102/00346543211042423/ASSET/IMAGES/LARGE/10.3102.00346543211042423-FIG2.JPG>.
- Sharma, H., Soetan, T., Farinloye, T., Mogaji, E., Noite, M.D.F., 2022. AI Adoption in Universities in Emerging Economies: Prospects, Challenges and Recommendations. *Re-Imagining Educational Futures in Developing Countries: Lessons from Global Health Crises*, pp. 159–174. https://doi.org/10.1007/978-3-030-88234-1_9/COVER.
- Sok, S., Heng, K., 2023. ChatGPT for education and research: a review of benefits and risks. *SSRN Electron. J.* <https://doi.org/10.2139/SSRN.4378735>.
- Stanca, L., Dabija, D.C., Păcurar, E., 2022. Community of practice: converting IT graduate students into specialists via professional knowledge sharing. *Kybernetes* 51 (2), 557–581. <https://doi.org/10.1108/K-10-2020-0711/FULL/HTML>.
- Staniskis, J.K., Katiliute, E., 2016. Complex evaluation of sustainability in engineering education: case & analysis. *J. Clean. Prod.* 120, 13–20. <https://doi.org/10.1016/J.JCLEPRO.2015.09.086>.
- Stephens, J.C., Hernandez, M.E., Román, M., Graham, A.C., Scholz, R.W., 2008. Higher education as a change agent for sustainability in different cultures and contexts. *Int. J. Sustain. High Educ.* 9 (3), 317–338. <https://doi.org/10.1108/14676370810885916/FULL/XML>.
- Tasdemir, C., Gazo, R., 2020. Integrating sustainability into higher education curriculum through a transdisciplinary perspective. *J. Clean. Prod.* 265, 121759. <https://doi.org/10.1016/J.JCLEPRO.2020.121759>.
- Tchamyou, V.S., 2020. Education, lifelong learning, inequality and financial access: evidence from African countries. *Contemporary Social Science* 15 (1), 7–25. <https://doi.org/10.1080/21582041.2018.1433314>.
- Thomas, J., Kneale, D., McKenzie, J.E., Brennan, S.E., Bhaumik, S., 2019. Determining the scope of the review and the questions it will address. *Cochrane Handbook for Systematic Reviews of Interventions* 13–31. <https://doi.org/10.1002/9781195366604.CH2>.
- Thorp, H.H., 2023. ChatGPT is fun, but not an author. *Science* 379 (6630), 313. <https://doi.org/10.1126/SCIENCE.ADG7879>.
- Tokayev, K.-J., 2023. Ethical implications of large language models A multidimensional exploration of societal, economic, and technical concerns. *Int. J. Soc. Analytics* 8 (9), 17–33. <https://norislab.com/index.php/ijsa/article/view/42>.
- Tomas, L., Girenti, S., Jackson, C., 2015. Pre-service teachers' attitudes toward education for sustainability and its relevance to their learning: implications for pedagogical practice, 23 (3), 324–347. <https://doi.org/10.1080/13504622.2015.1109065>.
- Tricco, A.C., Lillie, E., Zarin, W., O'Brien, K.K., Colquhoun, H., Levac, D., Moher, D., Peters, M.D.J., Horsley, T., Weeks, L., Hempel, S., Akl, E.A., Chang, C., McGowan, J., Stewart, L., Hartling, L., Aldcroft, A., Wilson, M.G., Garritty, C., et al., 2018. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann. Intern. Med.* 169 (7), 467–473. <https://doi.org/10.7326/M18-0850>.
- Valentin, Nagore, 2014. The sustainable university—a model for the sustainable organization. *Romania* 6 (2). <https://doi.org/10.1515/msd-2015-0002>.
- Van Poeck, K., König, A., Wals, A.E.J., 2018. Environmental and sustainability education in the Benelux countries: research, policy and practices at the intersection of education and societal transformation, 24 (9), 1234–1249. <https://doi.org/10.1080/13504622.2018.1477121>.
- Verma, P., Kumar, V., Daim, T., Sharma, N.K., Mittal, A., 2022. Identifying and prioritizing impediments of industry 4.0 to sustainable digital manufacturing: a mixed method approach. *J. Clean. Prod.* 356, 131639. <https://doi.org/10.1016/J.JCLEPRO.2022.131639>.
- Waas, T., Verbruggen, A., Wright, T., 2010. University research for sustainable development: definition and characteristics explored. *J. Clean. Prod.* 18 (7), 629–636. <https://doi.org/10.1016/J.JCLEPRO.2009.09.017>.
- Wals, A.E.J., 2008. From cosmetic Reform to Meaningful Integration: Implementing Education for Sustainable Development in Higher Education Institutes - the state of affairs in six European countries. *DHO.* <https://research.wur.nl/en/publication/s/from-cosmetic-reform-to-meaningful-integration-implementing-educa>.
- Wölfel, M., Taecharungroj, V., 2023. What Can ChatGPT Do? Analyzing early reactions to the innovative AI chatbot on Twitter. *Big Data Cognit. Comput.* 7 (1), 35. <https://doi.org/10.3390/BDC7010035>.
- Xu, L., Da, Xu, E.L., Li, L., 2018. Industry 4.0: state of the art and future trends. *Int. J. Prod. Res.* 56 (8), 2941–2962. <https://doi.org/10.1080/00207543.2018.1444806>.